Belief Disagreement and the Housing Credit Channel

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Abstract

When households disagree more about future house prices, lenders tighten credit. In survey, bank, and loan-level data, wider dispersion of house-price expectations is associated with tighter lending standards, lower loan-to-value ratios, and higher mortgage denial rates for low-income applicants. I develop a heterogeneous-agent model where homebuyers, some of whom are at their collateral constraint, hold heterogeneous beliefs about home prices, subject to a dispersion shock that widens the distribution *across* these beliefs. A disagreement-driven credit contraction moves aggregate consumption little but falls disproportionately on low-wealth borrowers. A monetary easing that offsets aggregate consumption restores little of their loss.

Keywords: belief disagreement, heterogeneous agents, cognitive discounting, stochastic volatility

JEL Classifications: D84, E44, G21, R31

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1 Introduction

The recent increase in house-price belief disagreement (see, e.g., Figure 1) has rekindled a productive discussion about its effects and, perhaps more importantly, who stands to lose the most when households disagree about future house prices. Yet clear answers remain elusive. On the one hand, the most optimistic borrowers bid up house prices (Burnside et al., 2016) and stand to lose the most when house prices fall. On the other hand, more pessimistic borrowers tend to choose higher leverage (Bailey et al., 2018), which exposes them to downside risk. When stratifying households by income and wealth instead of by their beliefs, it is also unclear who truly benefits from rising disagreement about house prices. In a belief-driven cycle, consumption declines the most among households with much of their wealth in housing (Kaplan et al., 2020), whereas middle- and high-income borrowers account for a rising share of mortgage defaults during crises (Adelino et al., 2016).

In this paper, I examine the heterogeneous effects of house-price belief disagreement and show that the negative effects tend to fall on marginal borrowers. A lender cannot reliably pin down borrowers' beliefs and therefore relies on information about the distribution of those beliefs to decide how much to tighten credit. Jointly with idiosyncratic productivity and the agent's own belief, this credit-tightening channel generates heterogeneous credit constraints. When beliefs spread out because of a disagreement shock, credit tightens, making the borrowing constraint bind for those who borrow up to it. Whereas the literature either focuses on disagreement between borrowers and lenders (Geanakoplos, 2010; Simsek, 2013) or allows disagreement to operate via the demand side (Burnside et al., 2016; Dong et al., 2025), here I study how disagreement *across* households about future house prices affects consumption heterogeneously via the credit channel, in which the lender tightens credit based on how dispersed such beliefs are.

Using the NY Fed Survey of Consumer Expectations (SCE), the Michigan Survey of Consumers, the Senior Loan Officer Opinion Survey (SLOOS), 30.4 million residential mortgage applications from HMDA, and 10,956 commercial mortgage loans from CMBS disclosures, I document three motivating facts.

1. Disagreement is associated with tighter lending standards: Lagged house-price belief disagreement is positively correlated with tighter bank lending standards (both residential and commercial).

2. Downside tail asymmetry: Credit tightening is more strongly correlated with disagreement in the pessimistic lower half of the forecast distribution than with disagreement in the optimistic upper half.

3. The associated rise in mortgage denials is largest for the lowest-income applicants: A one-standard-deviation rise in disagreement is associated with a 0.27 percentage point (pp) higher mortgage denial rate for applicants in the bottom income quartile, compared with 0.09 pp in the second quartile and 0.01 pp in the top quartile.

To quantify the macroeconomic and distributional implications of this channel, I develop a heterogeneous-agent model with collateral constraints, in which homebuyers have heterogeneous beliefs about home prices and face idiosyncratic income shocks. I then introduce a dispersion shock that widens the distribution *across* these beliefs, rather than the distribution of an individual agent's belief over time. The bank takes the distribution of beliefs into account when deciding how *tight* a homebuyer's collateral constraint should be. The rest of the model features a representative entrepreneur producing intermediate goods using commercial real estate, a continuum of monopolistically competitive retailers, a home-construction sector, a labor union, and a central bank. The model features sticky prices and real-wage rigidity.

Solving the model using the Reiter (2009) method, extended to second order as in Winberry (2018), and calibrating it to selected aggregate and survey moments, I find that belief disagreement is costly, perhaps especially for agents at the lower end of the wealth and income distribution. Specifically, a disagreement shock generates a cumulative consumption response of -18.5 percent for the poorest wealth quintile, compared with -0.8 percent for the richest. The 3% of households at the collateral constraint account for around 28% of the aggregate consumption decline on impact. Most importantly, these results operate through the credit channel, in which the financial intermediary prices borrowers' belief disagreement.

To check whether the disagreement shock is simply a manifestation of other standard shocks, I next consider how the model responds to a monetary tightening shock and an economic contraction in the form of a negative TFP shock. A monetary tightening that lowers aggregate consumption by the *same* amount is spread almost evenly across the wealth distribution. A negative TFP shock that lowers aggregate consumption by the *same* amount yields a consumption-response gap of around 2 percentage points between the top and bottom quintiles, whereas this number is around 18 percentage points for

disagreement shocks. Moreover, disagreement shocks account for more than 90% of the forecast-error variance of the consumption gap between the poorest and the richest quintiles, compared with around 45% for aggregate consumption.

I also find that the distributional results are not necessarily driven by how agents are sorted by their beliefs. Specifically, within income groups, optimists tend to bear more of the brunt of disagreement shocks, largely because more of them borrow up to the limit. This is where the credit channel comes into play: the lender takes into account the dispersion of borrowers' beliefs when setting the loan-to-value ratio, and this ratio tends to disproportionately affect marginal borrowers.

More broadly, this paper is related to four main strands of the literature. The first strand studies the effects of dispersion in beliefs. Previous work documents dispersion in survey data and links it to macroeconomic outcomes (see, e.g., Mankiw et al., 2003; Bachmann et al., 2013; Falck et al., 2021, among others). Coibion and Gorodnichenko (2012) show that the response of forecast disagreement to shocks helps distinguish between competing models of information rigidity. Notably, this line of work suggests that disagreement *across* agents is different from the agents' own subjective uncertainty (Lahiri and Sheng, 2010), which I illustrate in Figure 1. Relatedly, disagreement has been tied to asset prices and trading volume (Scheinkman and Xiong, 2003), and investors' beliefs are highly heterogeneous and are reflected in their portfolios and trades (Giglio et al., 2021). Guerreiro (2023) shows that disagreement across households amplifies business cycles through aggregate demand, and Dong et al. (2025) show that inflation disagreement weakens monetary transmission by pushing more households against their borrowing limits.

My contribution to this strand of the literature is to show who is affected by an increase in house-price belief disagreement and to provide a channel that explains the shock's distributional impact. Since a lender who cannot sort borrowers by their beliefs must take into account the entire distribution, a disagreement shock tightens credit for every borrower, with the burden falling squarely on those at the margin. In this sense, the lender's problem is one of forming a belief about the distribution of borrower beliefs, which is an example of the higher-order beliefs that the literature has considered (see, e.g., Coibion et al., 2021). Relatedly, Manove and Padilla (1999) and Exler et al. (2025) study lenders who cannot observe borrower optimism directly, and in Coval and Thakor (2005) a rational intermediary bridges optimistic entrepreneurs and pessimistic investors. Unlike these papers, the lender here prices the dispersion of beliefs, not their levels.

The second strand studies how housing collateral transmits shocks to key macroeconomic aggregates. This strand of the literature typically features housing collateral constraints in the tradition of Kiyotaki and Moore (1997) and Iacoviello (2005) (see, e.g., Iacoviello and Neri, 2010; Greenwald, 2018; Kaplan et al., 2020, among others). One key insight from this literature is that credit supply can play an important role in how leverage affects the real economy (see, e.g., Mian and Sufi, 2009, 2022; Favara and Imbs, 2015; Guerrieri and Lorenzoni, 2017; Justiniano et al., 2019; Sapci and Vu, 2022, among others).

This strand of the literature also asks how lenders ration credit (Stiglitz and Weiss, 1981) and how margins respond to the risk of collateral values (see, e.g., Brunnermeier and Pedersen, 2009; Adrian and Shin, 2014, among others). Jiang and Zhang (2025) show that houses with more dispersed values receive less credit. Diamond and Landvoigt (2022) let intermediaries price household leverage, and the risk shocks of Christiano et al. (2014) and Forlati and Lambertini (2011) tighten credit terms by raising the dispersion of idiosyncratic returns. The shock here is their belief analog. Most closely related to the disagreement side of my paper, Geanakoplos (2010) shows that leverage collapses when lenders become more uncertain about downside outcomes, and Simsek (2013) shows that disagreement about downside states, rather than upside states, tightens collateral constraints and limits leverage. Credit conditions, in turn, shape the cross-section of house prices (Landvoigt et al., 2015). To this strand of the literature, I contribute by documenting novel motivating facts regarding belief disagreement and credit tightening, and by showing how the effects of a belief disagreement shock propagate through the credit channel to the marginal borrower.

At the intersection of housing and beliefs is a third line of literature, which documents that homebuyers hold widely heterogeneous, experience-driven house-price beliefs and, more importantly, that these beliefs can alter homebuyers' decisions (see, e.g., Piazzesi and Schneider, 2009; Case et al., 2012; Armona et al., 2019; Bailey et al., 2019; Kuchler and Zafar, 2019, among others). I model the belief wedge as an idiosyncratic cognitive discount factor in the tradition of bounded-rationality macroeconomics (see, e.g., Gabaix, 2020; Angeletos and Huo, 2021, among others). Farhi and Werning (2019) show that bounded rationality and incomplete markets with occasionally binding constraints are complements in mitigating the effects of monetary policy. One source of novelty is that I let belief shocks, and not only productivity shocks, differ across households.

Last but not least, my paper is also related to the literature on incomplete markets and uninsurable risk à la Aiyagari (1994), and specifically to work on how borrowing constraints shape the transmis-

sion of shocks across heterogeneous households (see, e.g., Kaplan and Violante, 2014; Kaplan et al., 2018; Auclert, 2019; Bayer et al., 2024, among others). Methodologically, since I use the projection-and-perturbation approach of Reiter (2009) and Winberry (2018) and incorporate second-moment shocks, my paper is naturally related to the literature that builds on that method (see, e.g., Bayer and Luetticke, 2020; Auclert et al., 2021, among others) and the use of stochastic volatility in DSGE models (see, e.g., Fernández-Villaverde et al., 2015; Bloom et al., 2018, among others).

Among the more recent work, closest to this paper are three housing models that combine heterogeneous beliefs with credit limits.[1] Ludwig et al. (2024) feed the survey distribution of house-price expectations into a life-cycle model with loan-to-value and debt-to-income limits, Dong et al. (2022) let a belief cutoff pick the marginal buyer at a given loan-to-value ratio, and Tomura (2013) let optimistic borrowers face savers who do not share their optimism. None of these papers has a lender whose terms respond to the dispersion of borrower beliefs. Letting the lender's terms respond to dispersion connects the channel to Fact 2, since a lender that prices a low quantile of recovery is exposed to the pessimistic tail, and to Fact 3, since the tightening lands on borrowers at the limit.

The paper proceeds as follows. Section 2 documents three motivating facts on belief disagreement and credit tightening. Section 3 presents a dynamic model in which homebuyers have heterogeneous beliefs about home prices. Section 4 discusses the model's solution and calibration. Section 5 presents the model's results on how belief disagreement shocks affect key macroeconomic aggregates. Section 6 documents the model-implied distributional effects of these shocks and explains the mechanisms behind them. Section 7 concludes.

2 Disagreements in the Surveys

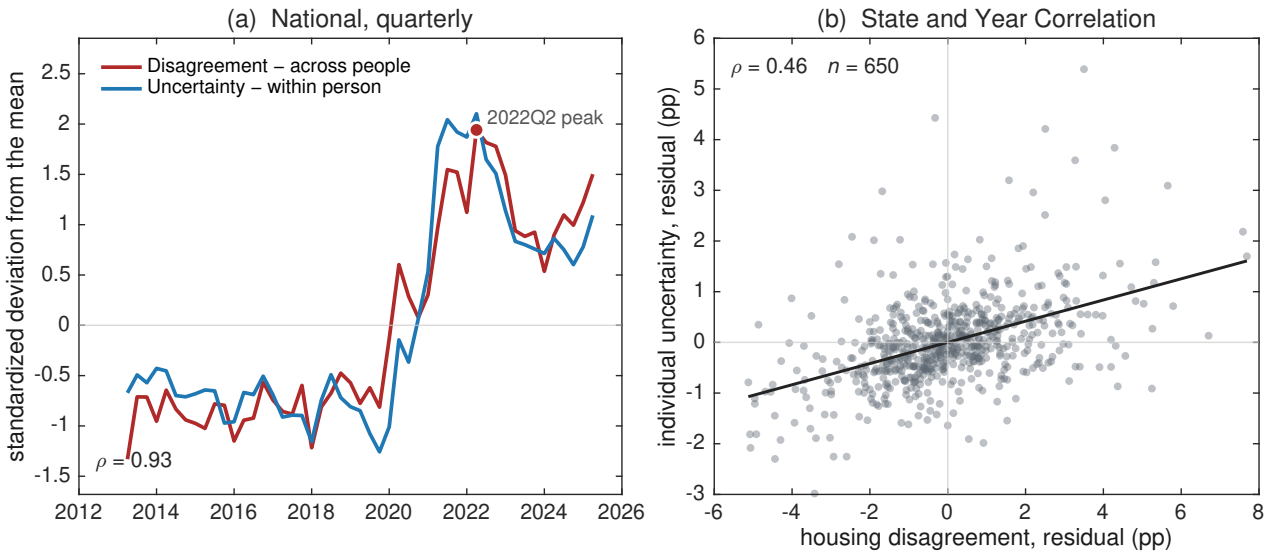
2.1 Disagreement vs. Uncertainty

Using data from the Federal Reserve Bank of New York's Survey of Consumer Expectations (SCE), Figure 1 plots the correlation between housing-price belief disagreement and uncertainty at the national level (Panel (a)) and, at the state level, in residuals after absorbing state and year fixed effects (Panel (b)).[2] Here, disagreement is the cross-sectional standard deviation of individual twelve-month home-price

[1] Table Appendix.1 in the Appendix provides a comparison with some of the closest antecedents of this paper. Besides the three papers listed, the remaining rows are the demand-side disagreement papers and the risk-shock and lender-leverage precedents discussed above.

[2] See Armantier et al. (2017) for a description of the SCE.

Figure 1: Housing Price Belief Disagreement and Uncertainty



Notes. Disagreement is the cross-sectional standard deviation of individual twelve-month home-price point forecasts (SCE Q31v2 winsorized at the 1st and 99th percentiles). Uncertainty is the mean width of each respondent's subjective density (mean C1 interquartile range). Panel (a) standardizes both national quarterly series on a single axis, 2013Q2–2025Q2, $n = 49$. The marked quarter is the sample maximum of disagreement. Panel (b) plots residuals of individual uncertainty and housing-belief disagreement after absorbing state and year fixed effects. Each point represents a state-year. Values for both series can be negative: in panel (a), series are standardized by subtracting the mean and dividing by the standard deviation. In panel (b), values are residuals after absorbing fixed effects.

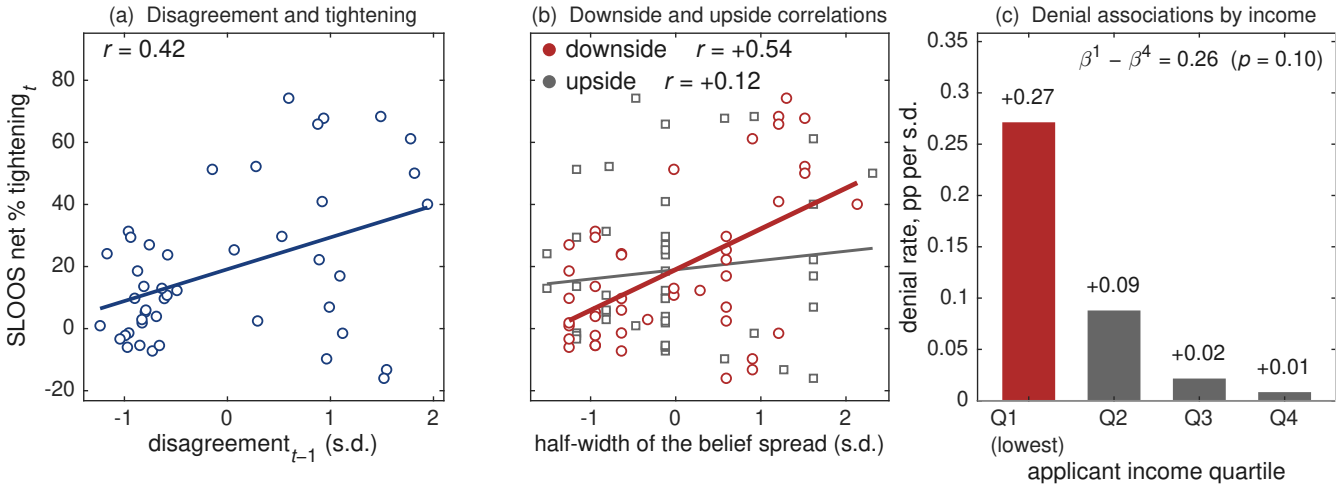
point forecasts, whereas uncertainty is the mean width of each respondent's subjective density.

Two implications stand out from Figure 1. First, disagreement is positively correlated with subjective uncertainty, and both jump after the onset of COVID-19. Second, disagreement is strongly correlated with uncertainty at the national level but much less so once state and year effects are absorbed. In other words, after removing these fixed effects, disagreement may carry information distinct from subjective uncertainty. For instance, if I disagree with my neighbor about what a house should be priced at, we both have beliefs about the price, but those beliefs differ. If neither of us is sure how to price the house, then both of us are subjectively uncertain. These situations are distinct, which may explain the drop in correlation in Figure 1 once fixed effects are absorbed.

2.2 Motivating Evidence

I now present three motivating facts based on the SCE, the Senior Loan Officer Opinion Survey on Bank Lending Practices (SLOOS), and public mortgage application data from the Consumer Financial Protection Bureau (CFPB). Figure 2 summarizes these facts.

Figure 2: Three Motivating Facts



Notes. Panel (a): standardized housing-belief disagreement D_{t-1} against SLOOS net commercial-real-estate tightening S_t, 2014Q1–2025Q1, $n = 45$. Panel (b): the same left-hand side against the two half-widths of the belief cross-section, $\text{DOWN}_t = p_{50,t} - p_{10,t}$ and $\text{UP}_t = p_{90,t} - p_{50,t}$, each standardized and measured from the median. Panel (c): the four β^q of Eqn. (1), in percentage points of the denial rate per standard deviation of state disagreement. The annotation in Panel (c) reports the difference between the bottom- and top-quartile coefficients, $\beta^1 - \beta^4$, with its clustered p-value. The text reports the gap between β^1 and the average of the other three quartiles.

Fact 1: Disagreement is positively correlated with credit tightening. Panel (a) plots one-quarter-lagged standardized housing-belief disagreement D_{t-1}, as computed in Figure 1, against credit tightening. I construct disagreement using the public core microdata from the SCE. The vertical axis denotes credit tightening S_t, defined as the simple average of three SLOOS series (*SUBLPDRCSC*, *SUBLPDRCSN*, and *SUBLPDRCSM*) downloaded from FRED.[3] Each point represents a national quarterly observation. Panel (a) shows that disagreement is positively correlated with credit tightening ($r = 0.42$).[4]

Fact 2: The correlation between disagreement and credit tightening is largely driven by the lower tail of the belief distribution. Panel (b) is a scatter plot of credit tightening (S_t) against the upper ($p_{90,t} - p_{50,t}$) and lower ($p_{50,t} - p_{10,t}$) halves of the belief spread (constructed from respondents' reported house-price forecasts). Each point represents a quarter. The positive correlation documented in Panel (a) is driven largely by disagreement in the lower tail. This observation dovetails with evidence that more

[3]Specifically, the three series are Net Percentage of Domestic Banks Tightening Standards for Commercial Real Estate Loans for Construction and Land Development Purposes (*SUBLPDRCSC*), Net Percentage of Domestic Banks Tightening Standards for Commercial Real Estate Loans Secured by Nonfarm Nonresidential Structures (*SUBLPDRCSN*), and Net Percentage of Domestic Banks Tightening Standards for Commercial Real Estate Loans Secured by Multifamily Residential Structures (*SUBLPDRCSM*).

[4]Additional evidence using loan-level data on commercial mortgages appears in Section 2.3 and Table 2. Overall, I find that disagreement is also positively correlated with credit tightening for commercial real estate loans at the loan level.

pessimistic homebuyers choose higher leverage (Bailey et al., 2019) and with a model in which disagreement about downside states, rather than upside states, tightens collateral constraints (Simsek, 2013). Pessimism about the state of the economy is also documented to be more prevalent among households with low income and education (Das et al., 2020). I note that Fact 2 does not answer the question of *who* (e.g., low- vs. high-income) is more pessimistic. In the SCE, respondents with household income below $50,000 disagree more than those above $100,000 but are, on average, more optimistic about house prices.

Fact 3: Credit tightening tends to fall on applicants with the lowest income. I estimate the following regression:

$$\text{Denial}^q_{s,y} = \alpha^q_s + \delta^q_y + \beta^q D_{s,y} + u^q_{s,y}, \qquad q = 1, \ldots, 4, \tag{1}$$

separately for each applicant-income quartile q, with state and year fixed effects and standard errors clustered by state. The data come from the public HMDA Loan/Application Register, drawn from the CFPB Data Browser and restricted to first-lien, owner-occupied home-purchase applications. The sample consists of 30.4 million applications from 2018 to 2024.

Turning to the details, I collapse the HMDA data by state and year and compute the denial rate as the share of applications that are denied. Specifically, $\text{Denial}^q_{s,y}$ is the share of quartile-q applications in state s and year y that are denied. The total number of applications includes those that are originated, approved but not accepted, or denied. Income quartiles are defined within each state-year, so $q = 1$ is low-income relative to that state's applicant pool in that year. α^q_s and δ^q_y denote state and year fixed effects. $D_{s,y}$ is standardized state-level disagreement (standardized across state-years). I estimate Eqn. (1) separately by quartile and plot β^q for $q = 1, 2, 3, 4$ in Panel (c). Panel (c) shows that the point estimates fall monotonically with income: $\beta^1 = 0.27$ (s.e. 0.18), $\beta^2 = 0.09$ (0.12), $\beta^3 = 0.02$ (0.08), and $\beta^4 = 0.01$ (0.05). None is statistically significant at the 5 percent level on its own. In a stacked regression, the bottom-quartile coefficient exceeds the average of the other three by 0.23 percentage points (s.e. 0.13, $p = 0.07$).

Robustness. Table 1 further breaks down these correlations by demographic groups and homeownership, checks whether they are robust when focusing on residential lending, excludes the 2020 observations, and tests whether the same results hold when using stock prices instead of house prices. Specifically, columns 1–2 report the simple correlation between X_{t-1} and S_t, and columns 3–6 report β_1 from

the following regression:

$$S_t = \alpha + \sum_{\ell=0}^{2} \beta_\ell D_{t-\ell} + \sum_{\ell=1}^{2} \gamma_\ell S_{t-\ell} + \varepsilon_t, \tag{2}$$

with X in place of D. Overall, I find that the positive correlation between belief disagreement and lending standards, and the asymmetry between the two halves of the disagreement distribution, hold across these alternative measures and subgroups, although the regression coefficient β_1 in columns 3–6 is statistically insignificant for every row except the stock-price placebo, whose full-sample coefficient loses significance once 2020 is excluded.[5]

[Table 1 here]

2.3 Evidence from Recorded Commercial Mortgage Leverage

Evidence supporting Fact 1 also appears in recorded financing outcomes. Table 2 uses 10,956 whole commercial mortgage loans from commercial mortgage-backed securities (CMBS) disclosures, grouped into 33 origination quarters over 2016Q4–2024Q4. A one-standard-deviation increase in lagged national housing-belief disagreement is associated with a 1.69 percentage-point decline in composition-adjusted securitization loan-to-value (LTV), with a Newey-West standard error of 0.24. The association remains negative when mean expectations or subjective uncertainty enter separately, when lagged LTV is included, in four-quarter differences, and when Michigan disagreement replaces the SCE measure. These results extend the motivating evidence to realized commercial mortgage leverage.

[Table 2 here]

3 Model

The economic environment consists of a continuum of households indexed by $i \in [0, 1]$, a representative entrepreneur producing intermediate goods using commercial real estate, a continuum of monopolistically competitive retailers, a home-construction sector, a labor union, and a central bank. The housing and collateral-constraint setup is motivated by Iacoviello (2005). The model features price stickiness and

[5] For brevity, I report the counterparts of Table 1 that use the Michigan survey instead of the SCE in Table Appendix.3 in the Appendix. The two sources generate similar correlations, which is unsurprising given that the two disagreement measures track each other closely (see, e.g., Figure Appendix.5 in the Appendix). The coefficient on lagged disagreement, β_1, is statistically insignificant for aggregate disagreement in both surveys. For the downside half, it is positive but insignificant in the SCE and negative in the Michigan data, significantly so when 2020 is excluded.

real-wage rigidity à la Blanchard and Gali (2007). One source of novelty is that belief dispersion is priced into credit limits.

3.1 Households

Preferences, budget, and the borrowing constraint. Household i values consumption C_{it} and housing services H_{it} (with weight $j > 0$) and dislikes work l_{it}:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta_h^t \left(\frac{C_{it}^{1-\sigma_c}}{1-\sigma_c} + j \frac{H_{it}^{1-\sigma_h}}{1-\sigma_h} - \psi \frac{l_{it}^{1+\xi}}{1+\xi} \right) \right], \tag{3}$$

subject to

$$C_{it} + q_t \big(H_{it} - (1-\delta_h) H_{it-1} \big) + B_{it} = \frac{R_{t-1} B_{it-1}}{\pi_t} + w_t e_{it} l_{it} + F_t, \tag{4}$$

where q_t is the real house price, B_{it} denotes real one-period bonds (with $B_{it} < 0$ denoting borrowing), w_t is the real wage per efficiency unit, e_{it} is idiosyncratic labor productivity, and F_t is a lump-sum rebate of retail and construction profits. Idiosyncratic productivity follows a stationary AR(1) in logs, $\log e_{it} = \rho_e \log e_{it-1} + \sigma_e \epsilon_{it}^e$, which I discretize using the Rouwenhorst (1995) method.

Borrowing is limited by a collateral constraint à la Kiyotaki and Moore (1997) and Iacoviello (2005),

$$B_{it} \geq -\bar{\phi}_t^{\text{eff}} q_t H_{it}, \tag{5}$$

in which $\bar{\phi}_t^{\text{eff}}$ is the lender-priced household loan-to-value limit defined in Eqn. (16) below. It equals the constant $\bar{\phi}$ in steady state. Since the union allocates a uniform workweek $l_{it} = L_{et}$, labor income is $w_t e_{it} L_{et}$.

Let λ_{it} and φ_{it} denote the Lagrange multipliers associated with Eqns. (4) and (5), respectively. The household first-order conditions are:

$$\text{Consumption:} \quad C_{it}^{-\sigma_c} = \lambda_{it}, \tag{6}$$

$$\text{Bonds:} \quad \lambda_{it} = \beta_h \mathbb{E}_t \left[\frac{R_t}{\pi_{t+1}} \lambda_{it+1} \right] + \varphi_{it}, \tag{7}$$

$$\text{Housing:} \quad j H_{it}^{-\sigma_h} + \beta_h m_{it} \mathbb{E}_t \left[\lambda_{it+1} (1-\delta_h) q_{t+1} \right] + \varphi_{it} \bar{\phi}_t^{\text{eff}} q_t = \lambda_{it} q_t. \tag{8}$$

Eqns. (6)–(8) hold together with the complementary-slackness conditions $\varphi_{it} \geq 0$ and $\varphi_{it} \big(B_{it} +$

$\bar{\phi}_t^{\text{eff}} q_t H_{it}\big) = 0$. As a result, the constraint may bind for some households and not for others.

Heterogeneous, cognitively discounted house-price beliefs. Following Gabaix (2020), I assume that households do not fully perceive the state of the economy when forecasting future house prices. Specifically, household i's perceived sale value is $m_{it}\mathbb{E}_t[q_{t+1}]$, where $m_{it} \in (0,1)$ is an idiosyncratic cognitive discount factor. When $m_{it} \to 1$, the model approaches a standard rational-expectations model. When $m_{it} < 1$, household i undervalues both the house price and the gain from holding housing. Whereas Gabaix (2020) applies discounting only to deviations from steady state, the discount here applies to the level of the expected resale value, so $m_{it} < 1$ lowers valuations even in steady state.

Rather than specifying an AR(1) process directly for m_{it}, I apply a logit transformation to an underlying belief process z_{it}, which ensures that m_{it} lies strictly between 0 and 1. The household-specific shocks to z_{it} are also motivated by evidence that idiosyncratic risk accounts for a large share of house-level price risk (Giacoletti, 2021). In addition, I find that this transformation makes the model more tractable because it rules out extrapolative beliefs. Specifically,

$$z_{it} = (1-\rho_m)\bar{z} + \rho_m z_{it-1} + \sigma_{mt}\varepsilon_{it}^m, \qquad m_{it} = \frac{1}{1+e^{-z_{it}}}, \quad z_{it} \equiv \text{logit}(m_{it}), \tag{9}$$

$$\log \sigma_{mt} = (1-\rho_{sm})\log \bar{\sigma}_m + \rho_{sm}\log \sigma_{mt-1} + \sigma_v \nu_t, \tag{10}$$

where the log-AR(1) process for dispersion σ_{mt} follows Fernández-Villaverde et al. (2015). Here, in Eqn. (10), ν_t is a disagreement shock that changes how dispersed households' beliefs are.

The idiosyncratic innovation ε_{it}^m is i.i.d. across households and over time, with mean zero and unit variance. These innovations average out across households, so the mean of z_{it} stays at $\bar{z}$ and a shock to ν_t moves only the dispersion of beliefs.[6] Subtracting the mean $\bar{z}$ from Eqn. (9) and defining the cross-sectional variance $V_t^{\text{cross}} \equiv \int (z_{it} - \bar{z})^2 \, di$ implies the following law of motion for belief dispersion.[7]

$$V_t^{\text{cross}} = \rho_m^2 V_{t-1}^{\text{cross}} + \sigma_{mt}^2. \tag{11}$$

[6] Because the logistic map is concave at $\bar{z} > 0$, the mean of m_{it} falls slightly, by at most 0.002 percent after a one-standard-deviation shock. With the lender's pricing switched off ($\chi_q = 0$), the same shock moves output by less than 0.001 percent.

[7] From $z_{it} - \bar{z} = \rho_m(z_{it-1} - \bar{z}) + \sigma_{mt}\varepsilon_{it}^m$, squaring both sides and integrating over i yields:

$$V_t^{\text{cross}} = \rho_m^2 \underbrace{\int (z_{it-1} - \bar{z})^2 di}_{=V_{t-1}^{\text{cross}}} + 2\rho_m\sigma_{mt} \underbrace{\int (z_{it-1} - \bar{z})\varepsilon_{it}^m di}_{=0} + \sigma_{mt}^2 \underbrace{\int (\varepsilon_{it}^m)^2 di}_{=1}.$$

3.2 Lender's Pricing of Beliefs

Consider a lender that lends against a house from which it can recover at most $\mathcal{C}_t$. The lender cannot screen borrowers by m_{it} at origination, so it must decide how much credit to offer based on the dispersion of collateral values rather than on any individual borrower's belief.[8] The lender recovers $\mathcal{R}_{t+1}\mathcal{C}_t$, where $\mathcal{R}_{t+1} \in (0,1)$, net of any costs of repossession and sale. The lender charges the gross loan rate R^L and funds itself at $R^f < R^L$, earning the margin $g \equiv 1 - R^f/R^L$ per unit promised. Given a promise to repay $\vartheta\mathcal{C}_t$, the lender receives

$$\min\{\vartheta\mathcal{C}_t, \mathcal{R}_{t+1}\mathcal{C}_t\},$$

which is the capped debt payoff à la Merton (1974), and owes $(1-g)\vartheta\mathcal{C}_t$. Intuitively, g is the share of each unit of promise that the lender keeps, after paying its funders, when the promise is paid in full.

The lender thus maximizes the expected profit per unit of $\mathcal{C}_t$:

$$\max_{0\le\vartheta\le 1} \underbrace{\mathbb{E}_t^{\text{lend}}\big[\min\{\vartheta, \mathcal{R}_{t+1}\}\big]}_{\text{Expected Receivables}} - \underbrace{(1-g)\vartheta}_{\text{Owed}} = \max_{0\le\vartheta\le 1} g\vartheta - \underbrace{\mathbb{E}_t^{\text{lend}}\big[\max\{\vartheta - \mathcal{R}_{t+1}, 0\}\big]}_{\text{Expected Credit Loss}}, \tag{12}$$

where the promise $\vartheta \equiv R^L\mathcal{B}_t/\mathcal{C}_t$ is the repayment due on loan $\mathcal{B}_t$ as a share of the maximum amount the lender can recover. The lender in Eqn. (12) chooses the largest promise ϑ that it can accept.

Intuitively, ϑ can be interpreted as a loan-to-value ratio measured at repayment. The first part of Eqn. (12) implies that the lender receives either the promise or the recovered value, whichever is lower. The second part of Eqn. (12) decomposes expected profits into the margin gain net of expected credit losses. The same capped payoff structure in Eqn. (12) also features in the credit surface model by Geanakoplos and Rappoport (2026), who find that uncertainty shocks lead to higher bond spreads and a steeper schedule of such spreads across loan-to-value ratios.[9] In Eqn. (12), however, the lender keeps the loan rate constant but changes the promise, and the dispersion comes from beliefs, not from the collateral price itself (see Eqn. 14).

A marginal unit of promise earns g but goes unpaid whenever the house falls short of the promise.

[8] See, for example, Brunnermeier and Pedersen (2009), who assume that financiers set each margin so that losses exceed it with a fixed probability (their equations (6) and (7)), so that haircuts move with collateral price volatility. Relatedly, Adrian and Shin (2014) find that intermediary leverage is negatively related to measured Value-at-Risk.

[9] In their model, the marginal expected payoff of a larger promise j is $1 - F(j)$ (their Lemma 1). Setting this quantity to the cost of funds R^f/R^L yields my optimal condition below in Eqn. (13).

As a result, the largest promise ϑ_t^* the lender will accept satisfies

$$F_{R,t}(\vartheta_t^*) = g, \tag{13}$$

where $F_{R,t}$ is the distribution function of $\mathcal{R}_{t+1}$ under the lender's beliefs. Here, $F_{R,t}(\vartheta_t^*)$ is the probability that the promise is not met. Eqn. (13) is the loan condition à la Freimer and Gordon (1965) and Jaffee and Modigliani (1969), where the bank chooses the loan amount that maximizes expected profits, taking the loan rate as given. It collects the smaller of the promised repayment and the borrower's outcome. In equilibrium, the probability of default equals the loan rate net of the cost of funds. I also note that since debt caps the lender's upside, an increase in the dispersion of recoveries (holding the mean fixed) lowers expected receipts (Stiglitz and Weiss, 1981). In this sense, the idea that cross-sectional dispersion, not just the mean, helps determine the optimal maximum promise (i.e., the loan-to-value ratio set by the lender) is motivated by the credit rationing literature.

3.3 Recovery Rate and Disagreement

In the case of late payment, the lender could (1) try to work out a settlement plan or (2) proceed with a forced sale of the property. In the first case, the borrower pays whatever the borrower believes the house is worth, namely m_{it}. Thus, in this case, the lender can recover at most m_{it}. In the second case, suppose the house is put on the market and another homebuyer $j \neq i$ purchases it, in which case the purchase price is whatever the buyer values it at, namely m_{jt}.

Either way, from the lender's perspective, the recovery rate $\mathcal{R}_{t+1}$ is drawn from the cross-sectional distribution of beliefs m_{it}, with mean $\bar{z}$ and variance $s_t^2 \equiv \rho_m^2 V_t^{\text{cross}} + \mathbb{E}_t \sigma_{mt+1}^2$. To connect recovery risk to disagreement while maintaining tractability, I use $\sqrt{V_t^{\text{cross}}}$ to approximate s_t.[10] Intuitively, this implies that recoveries are more dispersed when borrowers disagree more,

$$\operatorname{logit} \mathcal{R}_{t+1} = \bar{z} + \kappa_R \sqrt{V_t^{\text{cross}}} \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim N(0,1), \qquad \kappa_R > 0. \tag{14}$$

The logit transformation keeps the recovery rate between zero and one, as in Düllmann and Trapp (2004) and Chava et al. (2011). Equation (14) implies that recovery risk is proportional to disagreement, consistent with a collateral-recovery channel in the spirit of Jiang and Zhang (2025), who document that

[10] Solving the model with s_t instead of $\sqrt{V_t^{\text{cross}}}$ yields virtually identical distributional results. Indeed, the two series have a correlation of 0.997 in simulation.

lenders extend less credit against more dispersed collateral values. Here, instead of house-level price dispersion, I use belief dispersion.

Combining Eqn. (13) and Eqn. (14) yields

$$\operatorname{logit} \vartheta_t^* = \bar{z} - \underbrace{\kappa_R \Phi^{-1}(1-g)}_{\chi_q} \sqrt{V_t^{\text{cross}}},$$

or, letting $\chi_q \equiv \kappa_R \Phi^{-1}(1-g)$,

$$\operatorname{logit} \vartheta_t^* = \bar{z} - \chi_q \sqrt{V_t^{\text{cross}}}, \tag{15}$$

where Φ denotes the standard normal distribution function. Intuitively, Eqn. (15) implies that higher disagreement lowers the promise, holding the shortfall probability fixed at g. Here, ϑ_t^* is the share of a unit of collateral value that the lender is willing to lend against. Combining Eqns. (9) and (15) yields

$$\vartheta_t^* = \frac{1}{1+\exp\left(-(\bar{z} - \chi_q \sqrt{V_t^{\text{cross}}})\right)}, \qquad \text{and} \qquad \bar{\phi}_t^{\text{eff}} = \bar{\phi} \frac{\vartheta_t^*}{\vartheta_{ss}^*}, \tag{16}$$

where χ_q governs how the lender accounts for borrowers' belief dispersion. A higher χ_q implies a larger negative effect of belief dispersion on the lender's valuation of collateral (i.e., a lower ϑ_t^*). Here, ϑ_{ss}^* is the steady-state value of ϑ_t^*. Since the promise dictates what the lender can recover, ϑ_t^* effectively corresponds to a low quantile of next period's valuation, moving with the pessimistic tail of beliefs, which is consistent with Fact 2 reported in Section 2. Pricing a low quantile in this way follows the spirit of Simsek (2013), in whose model downside states are effectively priced by pessimistic lenders. Normalizing $g = 1\%$ implies $\kappa_R = 3.69$, which corresponds to recovery risk that is about 3.7 times as dispersed (in logit units) as homebuyers' beliefs.

To maintain tractability, I assume that in equilibrium the lender only sets the loan-to-value limit. Households and the entrepreneur borrow and save at the policy rate R_t, with the lender passing funds through. No borrower defaults, so there are no credit losses or intermediary profits to distribute.

3.4 The Rest of the Economy

The entrepreneur produces intermediate goods using commercial real estate H_{et-1} and labor,

$$Y_t = A_t H_{et-1}^{\alpha} L_{et}^{1-\alpha}, \qquad \log A_t = \rho_A \log A_{t-1} + \zeta_A \left(\log Y_{t-1} - \log \bar{Y}\right) + \sigma_A \varepsilon_t^A, \tag{17}$$

where $\zeta_A \geq 0$ captures feedback from aggregate demand that generates productivity gains in the spirit of Comin and Gertler (2006) and Anzoategui et al. (2019). The entrepreneur maximizes

$$\mathbb{E}_t \sum_s \beta_e^s \frac{C_{et+s}^{1-\sigma_{ce}}}{1-\sigma_{ce}},$$

with $\beta_e < \beta_h$, subject to

$$C_{et} + q_t\big(H_{et} - (1-\delta_h)H_{et-1}\big) + B_{et} = \frac{Y_t}{X_t} - w_t L_{et} + \frac{R_{t-1}B_{et-1}}{\pi_t}, \tag{18}$$

with X_t the price markup, and a binding collateral constraint

$$B_{et} \geq -\Xi_{e,t}^{\text{eff}}(1-\delta_h)q_t H_{et}\frac{\mathbb{E}_t[\pi_{t+1}]}{R_t}, \qquad \text{and} \qquad \Xi_{e,t}^{\text{eff}} = \Xi_e \frac{\vartheta_t^*}{\vartheta_{ss}^*}, \tag{19}$$

where $\Xi_{e,t}^{\text{eff}}$ denotes the lender's loan-to-value ratio, which equals the constant Ξ_e in steady state, and ϑ_t^* is given in Eqn. (16). The factor $(1-\delta_h)$ depreciates repossession proceeds, and $\mathbb{E}_t[\pi_{t+1}]/R_t$ discounts them to the origination date (Iacoviello, 2005). The entrepreneur is less patient than households ($\beta_e < \beta_h$), and households' precautionary saving keeps $\beta_h R/\pi < 1$ in the steady state, so $\beta_e R/\pi < 1$ and the entrepreneur borrows up to the limit. Following Iacoviello (2005), I assume that the constraint also binds in the steady state, and hence the parameter choices described in Section 4.2.

Following Iacoviello (2005), I assume Calvo (1983) pricing for retailers. Retailers buy intermediate goods competitively and sell differentiated varieties with elasticity of substitution θ and reset probability $1-\gamma$. To keep the model tractable and avoid the question of whose discount factor to use, I assume that retailers discount profits with the mass-weighted household stochastic discount factor (SDF), $\Lambda_{t,t+s} = \beta_h^s \bar{\lambda}_{t+s}/\bar{\lambda}_t$, where $\bar{\lambda}_t \equiv \int \lambda_{it}\, di$. In words, retailers share the same SDF, and the problem collapses to a standard sticky-price setup (see, e.g., Blanchard and Gali, 2007 or Sims, 2024 for further discussion).

Homebuilders build new housing subject to level and change-in-investment adjustment costs à la Topel and Rosen (1988) and Christiano et al. (2005). Specifically,

$$q_t = 1 + \chi_h \frac{I_t}{H_t} + \chi_{\Delta I}\frac{I_t - I_{t-1}}{H_t} - \beta_h \chi_{\Delta I}\mathbb{E}_t\left[\frac{I_{t+1}-I_t}{H_{t+1}}\right], \tag{20}$$

with $I_t = H_t - (1-\delta_h)H_{t-1}$ and $H_t = H_{et} + \int H_{it}\, di$. Here, χ_h and $\chi_{\Delta I}$ govern the level and change-in-investment adjustment costs, respectively.

I assume that there is a labor union that runs the supply side of the labor market. The union allocates labor hours $l_{it} = L_{et}$, which depend on the firm's labor demand at the market-prevailing wage

$$w_t = (1 - \alpha) Y_t / (L_{et} X_t),$$

and the notional wage is the membership's aggregate marginal rate of substitution, $w_t^{\text{flex}} = \bar{\psi} L_{et}^{\xi} / \bar{\lambda}_t$. Following Blanchard and Gali (2007), I assume the actual wage adjusts partially:

$$\log w_t = \gamma_w \log w_{t-1} + (1 - \gamma_w) \log w_t^{\text{flex}}, \tag{21}$$

where Eqn. (21) is a parsimonious, reduced-form way of capturing the slow adjustment of real wages. Blanchard and Gali (2007) show that an explicit model of staggered real-wage setting yields the same basic conclusions.

3.5 Monetary Policy and Market Clearing

I assume a Taylor (1993) rule with inertia,

$$R_t = R_{t-1}^{\rho_R} \left[R \left(Y_t / Y \right)^{\phi_Y} \left(\pi_t / \pi \right)^{\phi_\pi} \right]^{1-\rho_R} \exp(\varepsilon_t^R). \tag{22}$$

Markets clear. Goods: $Y_t = \int C_{it}\, di + C_{et} + I_t +$ adjustment costs. Labor: the union allocation implies $\int e_{it} l_{it}\, di = L_{et}$. Housing: $H_t = H_{et} + \int H_{it}\, di$. Bonds: $\int B_{it}\, di + B_{et} = 0$.

4 Solution and Calibration

4.1 Solution

The individual state is $\mathbf{s}_{it} = (x_{it-1}, m_{it}, e_{it})$, and the aggregate state consists of the cross-sectional distribution $\mu_t(\mathbf{s}_{it})$ together with $(A_t, \sigma_{mt}, V_t^{\text{cross}}, R_{t-1}, \dots)$. Here, $x_{it} = R_{t-1} B_{it-1} / \pi_t + q_t (1 - \delta_h) H_{it-1} + w_t e_{it} l_{it} + F_t$.[11] The model's equilibrium includes a set of policy functions, prices, and a law of motion $\mu_{t+1} = \Gamma(\mu_t; \mathbf{S}_t)$. All agents optimize, and all markets clear.

[11] The two predetermined stocks B_{it-1} and H_{it-1} enter only through the budget constraint, and only through the sum x_{it}, so wealth is sufficient for the pair. The borrowing constraint (5) is written on the *current* stock, and the labor condition is also in the current period.

I solve the model using the Reiter (2009) method, extended to second order à la Winberry (2018). This involves solving the household problem on a discrete grid, carrying the histogram over that grid as a vector of endogenous states, and perturbing around the stationary equilibrium. The individual state $\mathbf{s}_{it}$ is discretized using 8 grid points for x_{it-1}, 3 for m_{it}, and 3 for e_{it}, resulting in 225 predetermined variables per period (including bond-policy lags and other aggregate state variables). The perturbation is taken to second order with pruning.[12]

The steady-state household problem imposes the collateral constraint as an inequality, so it binds for some households and not for others. The perturbation holds this split fixed: bins at the limit in the steady state stay at the limit, and all others stay unconstrained. No bin crosses the limit along the impulse-response paths reported throughout the paper.

4.2 Calibration

I discipline the model using parameter values from the literature whenever possible. Table 3 lists all parameters and their values, along with sources where appropriate. For parameters without clear counterparts in the literature, I either estimate them from the data (e.g., χ_q) or choose values so that the model matches selected aggregate moments (e.g., σ_h and σ_{ce}).

[Table 3 here]

[Table 4 here]

Households. These parameters are mostly standard, borrowing heavily from Iacoviello (2005). The subjective discount factor β_h is 0.99, which corresponds to a discount rate of 4% annualized. The housing preference weight j is 0.1. I set σ_c to 2, the upper bound on relative risk aversion implied by the labor-supply evidence in Chetty (2006). The inverse Frisch elasticity is 2, following Chetty (2012). The household LTV limit is 0.8, a mortgage-market convention (conventional mortgages with an LTV above 80% generally require private mortgage insurance).

Lenders' dispersion multiplier χ_q. The parameter χ_q governs how cross-sectional belief dispersion (V_t^{cross}) translates into how much the lender will lend against the home (ϑ_t^*). Let $s_V = \text{sd}(\sqrt{V^{\text{cross}}})$ be the

[12] For robustness, I also solve the model at third order, which is considerably more costly than at second order. Figure Appendix.6 compares key impulse responses of aggregate variables to the disagreement shock at second and third order. Figure Appendix.7 presents a similar comparison across wealth quintiles.

simulated standard deviation of $\sqrt{V_t^{\text{cross}}}$ over time (not across i) implied by my calibration in Table 3. My starting point is the model-implied marginal effect of disagreement on lending standards,

$$\frac{d\log\bar{\phi}_t^{\text{eff}}}{d\tilde{D}_t} = \frac{d\log\vartheta_t^*}{d\tilde{D}_t} = -\chi_q(1-\vartheta_{ss}^*)s_V, \qquad \text{and} \qquad \tilde{D}_t \equiv \frac{\sqrt{V_t^{\text{cross}}} - \sqrt{V_{ss}^{\text{cross}}}}{s_V}, \tag{23}$$

where $\tilde{D}_t$ is the normalized model-side disagreement measure. Intuitively, Eqn. (23) is the semi-elasticity of the credit limit $\bar{\phi}_t^{\text{eff}}$ with respect to the model's belief dispersion. This gives a natural link between the model and the data:

$$-\frac{\hat{\beta}_1}{100} = \frac{d\log\bar{\phi}_t^{\text{eff}}}{d\tilde{D}_t} = -\chi_q(1-\vartheta_{ss}^*)s_V, \tag{24}$$

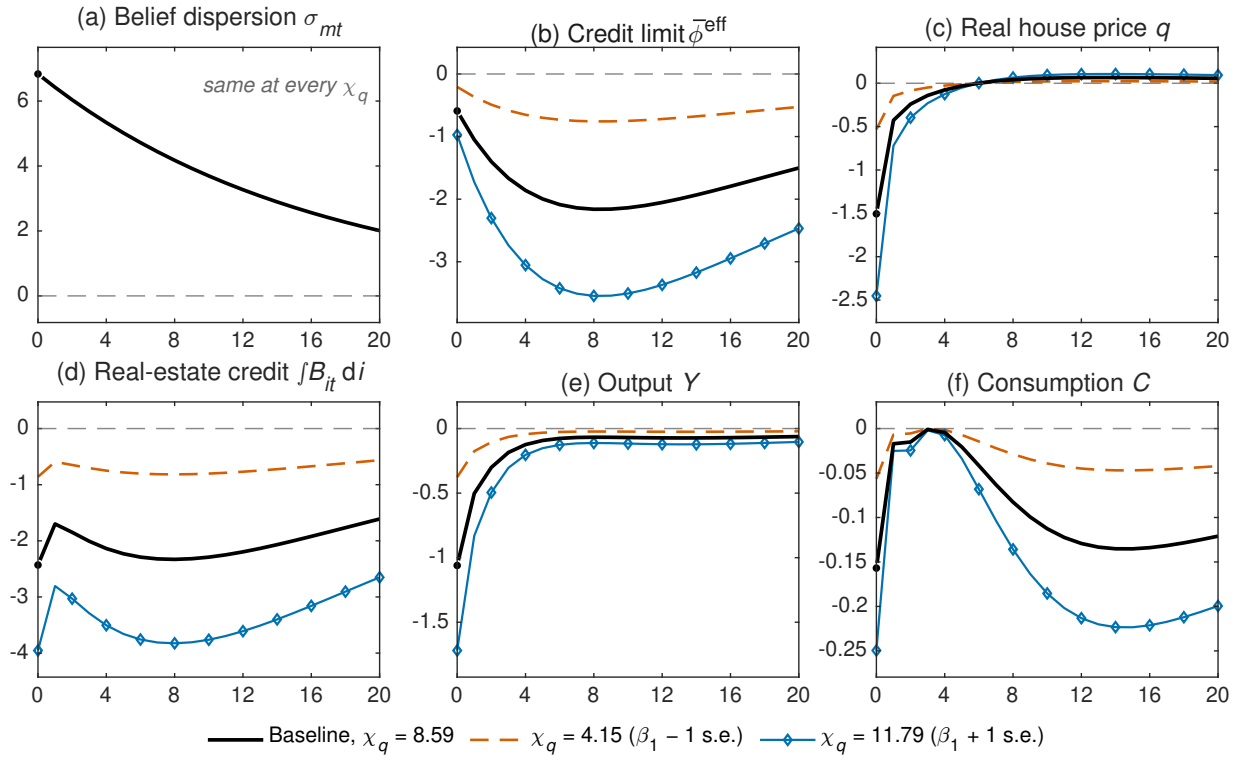
where $\hat{\beta}_1 = 9.54$ is the full-sample estimate of β_1 in Eqn. (2) for commercial real-estate lending standards (first row of Table 1). This implies $\chi_q = 8.59$.

Other parameters. The entrepreneur's subjective discount factor is 0.98, following Iacoviello (2005). Its CRRA parameter σ_{ce} is 2. The real-estate share α is 0.33, and the entrepreneur's LTV limit Ξ_e is 10%. I set the Calvo parameter γ to 0.75 and the elasticity of substitution θ to 5, which implies a markup of 25%. Motivated by Blanchard and Gali (2007), I set $\gamma_w = 0.9$. Housing depreciation δ_h is set to 0.025. I choose $\bar{m} = 0.75$ and $\rho_m = 0.92$, following Vu (2026), where I estimate the average cognitive discount and its persistence from the data. The value of $\bar{m}$ is also close to those used or estimated in the literature (e.g., 0.85 in Gabaix, 2020 and posterior means of 0.62–0.73 in Meggiorini, 2023). The Taylor-rule parameters (ρ_R, ϕ_Y, and ϕ_π) are chosen based on the recent literature (see, e.g., Carvalho et al. 2021 for discussion). I set the inflation target π to $1.02^{1/4}$ to reflect a 2% annual target.

4.3 Model versus Data: Untargeted Moments

I compare the standard deviations (Panel A), comovement (Panel B), persistence (Panel C), and cross-correlations (Panel D) of key macroeconomic aggregates between the model and the data in Table 4. Panel E reports the correlations of house prices and consumption with the bottom-50% net-worth share. The second column shows the data moments, whereas the third column reports the model counterparts (values in brackets report the 5–95% range across 1,000 simulated samples). Overall, I find that the model-generated moments are largely consistent with those in the data. For brevity, I relegate the detailed construction of these moments and the underlying series to Panel B of Table Appendix.2.

Figure 3: Aggregate Impulse Responses to a Belief Disagreement Shock



Notes. Impulse responses to a one-standard-deviation innovation ν_t to (10), simulated at order 2 with pruning. Vertical axes are in percent, horizontal axes in quarters since the impulse. The black solid line is the baseline $\chi_q = 8.59$, the dashed line $\chi_q = 4.15$ and the blue diamond-marked line $\chi_q = 11.79$.

5 Aggregate Responses to Belief Disagreement Shocks

5.1 Responses to Belief Disagreement

Figure 3 plots impulse responses to a one-standard-deviation dispersion shock—the innovation ν_t in Eqn. (10). I simulate the model with a one-standard-deviation shock to ν_t and report the difference relative to a simulation without the shock. Going forward, I refer to ν_t as a belief-disagreement shock. Responses are expressed as percentage deviations from the no-shock path. Figure 3 also reports results for three values of χ_q: the baseline ($\chi_q = 8.59$) and two alternatives computed using $\hat{\beta}_1 \pm$ one standard error, as described in Eqn. (24).

One key insight from Figure 3 is that output Y and consumption C both decline following a belief-disagreement shock, with output falling by about 1 percent in the baseline. Intuitively, these declines are driven by tighter credit limits $\bar{\phi}^{\text{eff}}$, followed by declines in the real house price q and real-estate

credit $\int B_{it}\, di$. The consumption response is particularly notable: it falls on impact, recovers by the third quarter, and then declines a second time, to about -0.14 percent. This pattern reflects the hump-shaped response of the credit limit that precedes the decline in consumption. More broadly, the decline in consumption is consistent with evidence that uncertainty depresses consumption (see, e.g., Coibion et al., 2024).

Response magnitudes for aggregate quantities are either small (e.g., consumption) or short-lived (e.g., output and the real house price), yet credit responses are persistent. This suggests that the mechanism operates primarily through the distribution across agents rather than through aggregate dynamics, which I explore in Section 6.1.

5.2 Aggregate Responses: Model and Data

I next check whether the model responses are consistent with the data. To that end, I simulate data from the model and construct data counterparts from FRED. Table Appendix.2 in the Appendix documents the sources and details the construction of the variables and their model equivalents.

Using both the model-simulated series and the data counterparts, I estimate local projections à la Jordà (2005). Specifically, for each outcome x and horizon $h = 0, \ldots, 8$, I consider the following local projection (LP) regression:

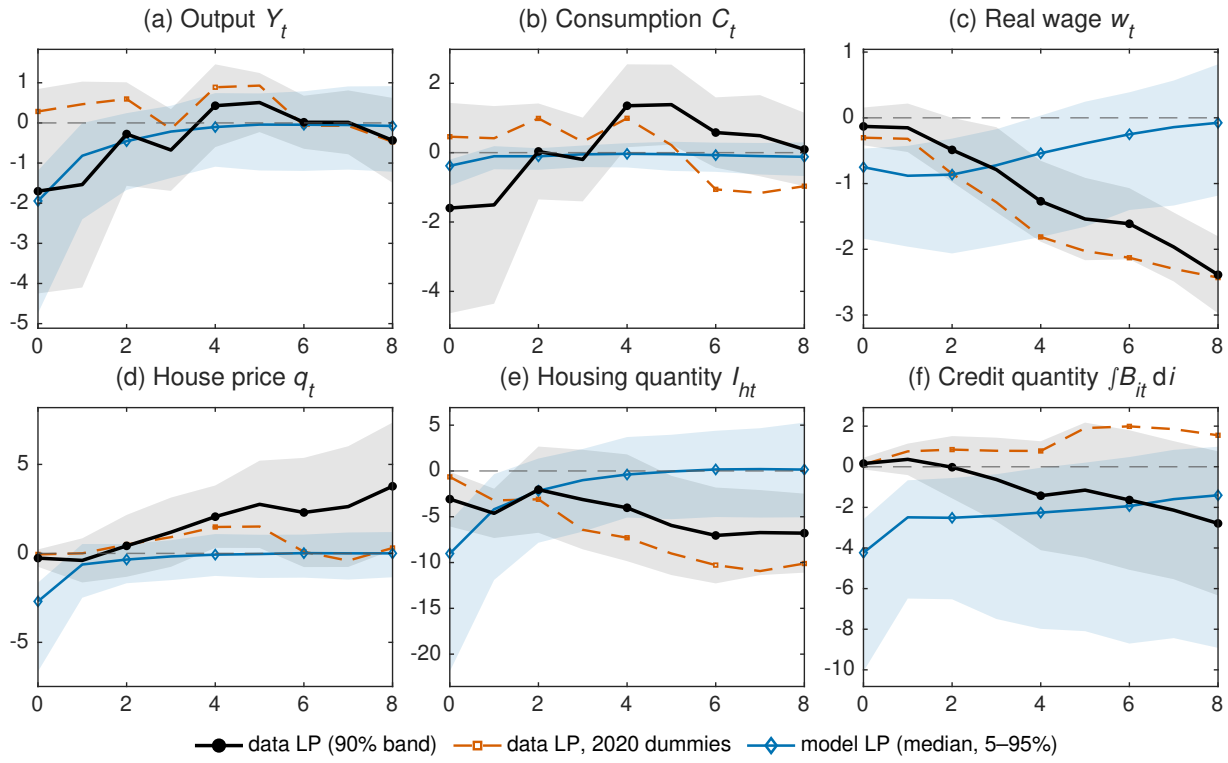
$$\sum_{s=0}^{h} \Delta \log x_{t+s} = \alpha_h + \beta_h u_t + \sum_{\ell=1}^{2} \gamma_{h\ell} u_{t-\ell} + \sum_{\ell=1}^{2} \delta_{h\ell} \Delta \log x_{t-\ell} + \varepsilon_{t+h}, \tag{25}$$

where u is standardized national disagreement. Note that u is the persistent *level* of dispersion, the empirical analog of σ_{mt}, not an identified structural shock.

Figure 4 plots impulse responses constructed from the local projections in Eqn. (25). I consider three response sets for six variables: output (Y_t), consumption (C_t), real wage (w_t), house price (q_t), housing quantity (I_{ht}), and credit quantity ($\int B_{it}\, di$). The first set uses the data in Table Appendix.2 (solid lines with circle markers). The second uses the same data but controls for COVID-19 periods (dashed lines with square markers). The third uses model-simulated data.

Figure 4 shows that the model responses of output, consumption, real wages, housing quantity, and credit quantity reasonably match the signs (largely negative) of the data responses and, to a degree, their shapes and magnitudes. House price responses are harder to match, with the model response hovering around zero while the data response drifts upward. Zooming in, the responses match less well when

Figure 4: Data versus model responses



Notes. I estimate Eqn. (25) on 1,000 simulated 46-quarter samples of the model (simulated at order 2, as elsewhere in the paper) and on the data. I plot the model median with its 5–95% range against the data estimate with a 90% Newey–West confidence band. Vertical axes are β_h in percent; horizontal axes are horizons in quarters. Table Appendix.2 in the Appendix details the construction of these variables.

2020 dummies are included, but overall, the model responses are broadly consistent with those in the data.

5.3 Counterfactual: Model, Data, and the Credit Channel

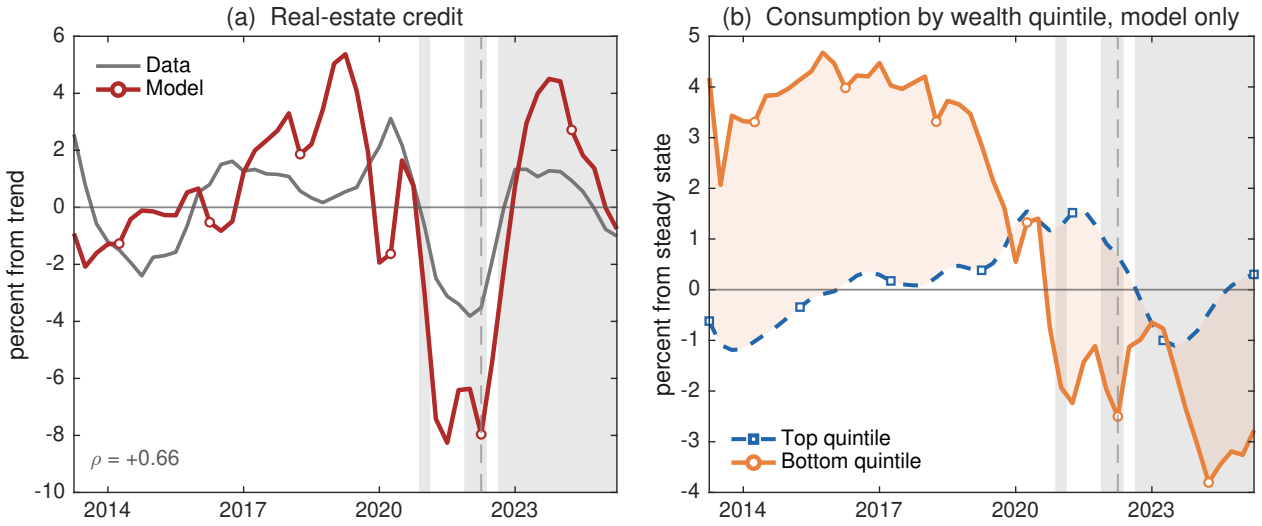
I consider a counterfactual exercise in which I back out the shocks that minimize the distance between the model-implied disagreement $\tilde{D}_t(\nu)$ and its empirical counterpart D_t. Specifically, I extract the historical shock sequence $\hat{\nu}$ as

$$\hat{\nu} = \arg\min_{\nu} \frac{1}{\sigma_D^2} \sum_t \left(\tilde{D}_t(\nu) - D_t\right)^2 + \sum_t \nu_t^2, \tag{26}$$

where σ_D scales the measurement error in $D_t = \tilde{D}_t(\nu) + \varepsilon_t^D$, $\varepsilon_t^D \sim N(0, \sigma_D^2)$, so the extracted shocks carry the same unit variance as in the calibration.

Figure 5 shows that disagreement shocks extracted from the data allow the model to track the path of

Figure 5: Counterfactual: Model vs. Data



Notes. The measured SCE disagreement path of Section 2.2, inverted onto ν_t by (26) with the productivity and policy shocks off. The dashed line is 2022Q2, where the survey peaks. *Panel (a)* plots real-estate loans at all commercial banks, deflated and entering as 100 log, against the model's real-estate credit. Both sides are HP filtered at $\lambda = 1600$ over the forty-nine quarters plotted, which are the quarters used for the annotated correlation ρ. *Panel (b)* plots the model's consumption of the bottom and top steady-state wealth quintiles as percent deviations from a constant steady state. In the shaded quarters, the simulated path puts some households above their borrowing limit (Section 4.1).

real-estate credit reasonably well (Panel (a)). The model overshoots the COVID-19 episode and generates more volatility than the data. Turning to Panel (b), consumption in the bottom wealth quintile tracks real-estate credit in Panel (a) much more closely than consumption in the top quintile. This is consistent with the result that households at the lower end of the wealth distribution bear the brunt of the shock through the credit channel.

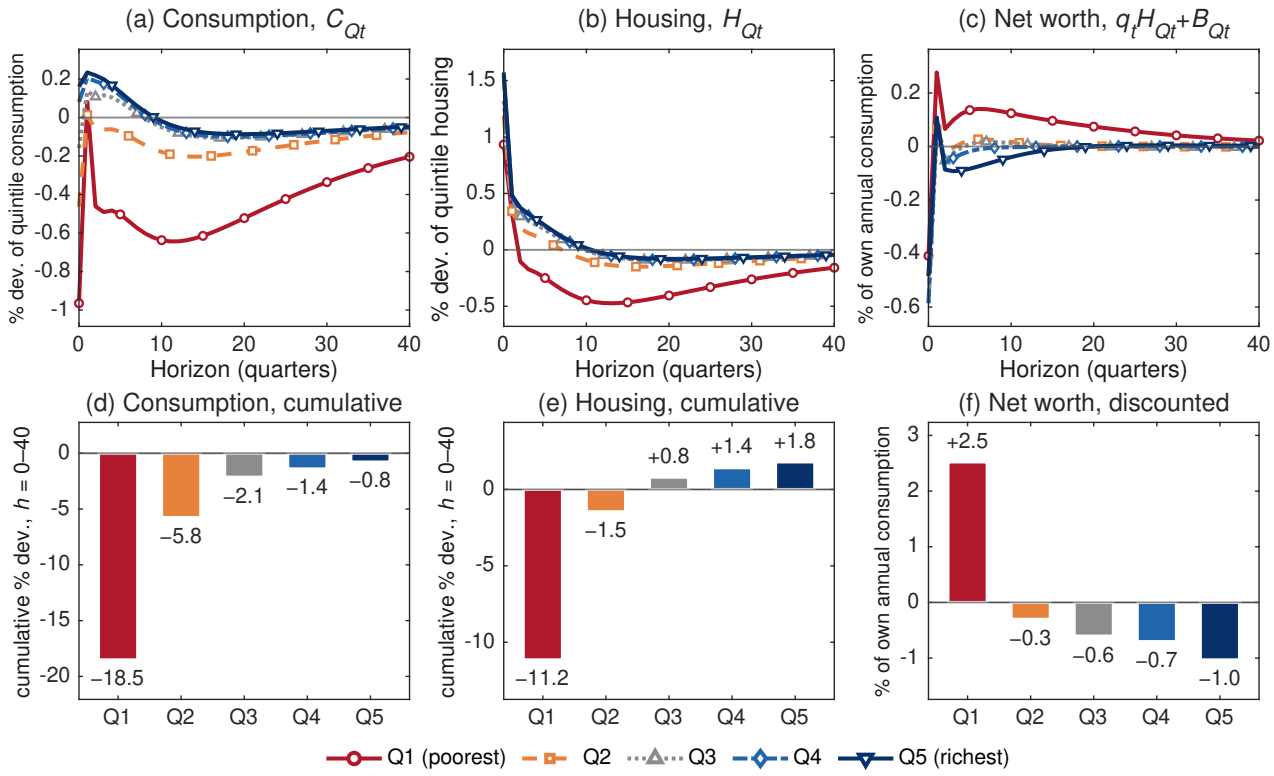
As a robustness exercise, Figure Appendix.4 in the Appendix repeats the counterfactual using the Michigan series instead of the SCE to extract shocks. The two sources produce similar counterfactual paths, which is unsurprising given that the series track each other closely (Figure Appendix.5).

6 Distributional Effects of Belief Disagreement Shocks

6.1 The Distributional Effects of Belief Disagreement

To understand the effects of belief disagreement across different agents, I plot, in Figure 6, the responses of consumption (C_{Qt}), housing (H_{Qt}), and net worth ($q_t H_{Qt} + B_{Qt}$) across wealth quintiles for each period after the shock (panels a, b, and c) and cumulated over the 40-quarter horizon (panels d and e) or, for net

Figure 6: Consumption, housing, and net worth by steady-state wealth quintile



Notes. Consumption, housing, and net worth by steady-state wealth quintile, one margin per column. The top row, panels (a) to (c), is the response to a $+1$ standard deviation impulse over $h = 0$ to 40. The bottom row, panels (d) to (f), reduces the path directly above it to one number per quintile. Quintiles are cumulative steady-state mass sorted by realized bond holdings.

worth, discounted over the full path (panel f). Here X_{Qt} is the mass-weighted quantity across all bins in quintile Q. In particular, each series is the percent deviation of the quintile aggregate, weighted by steady-state mass and levels and divided by that quintile's own steady-state total. Net worth is mass-weighted to a quintile mean and divided by four times that quintile's own steady-state consumption, so panels (c) and (f) are in percent of annualized consumption. Here, I sort households by their realized steady-state wealth position, measured by carried-in bond holdings b_k.[13]

Turning to the details in Figure 6, Panel (a) shows that consumption C_{Qt} falls the most for individuals in the lowest wealth quintile. In addition, the size of the fall lines up with the wealth ranking. Indeed, Panel (d), which plots the cumulative change in consumption for these five quintiles, also shows that individuals in Q1 experience the sharpest declines following an unexpected belief disagreement shock.

[13]Quintiles cannot be formed cleanly based on income because I use 3 bins to model e_{it}, so the cuts fall inside cells where households have identical income. For robustness, Figure Appendix.3 in the Appendix repeats Figure 6 with the distribution broken down by the three income bins.

Panels (b) and (e) tell the same story for housing services. Specifically, after a brief initial increase in housing for all quintiles, individuals at the lower end of the wealth distribution suffer the largest declines in housing services. Since consumption and housing services both decline for households at the lower end of the wealth distribution, they are able to save a bit more *relative* to their own steady-state consumption. As a result, after falling on impact, their net worth rises above its no-shock path and stays there for most of the horizon, as shown in Panels (c) and (f).

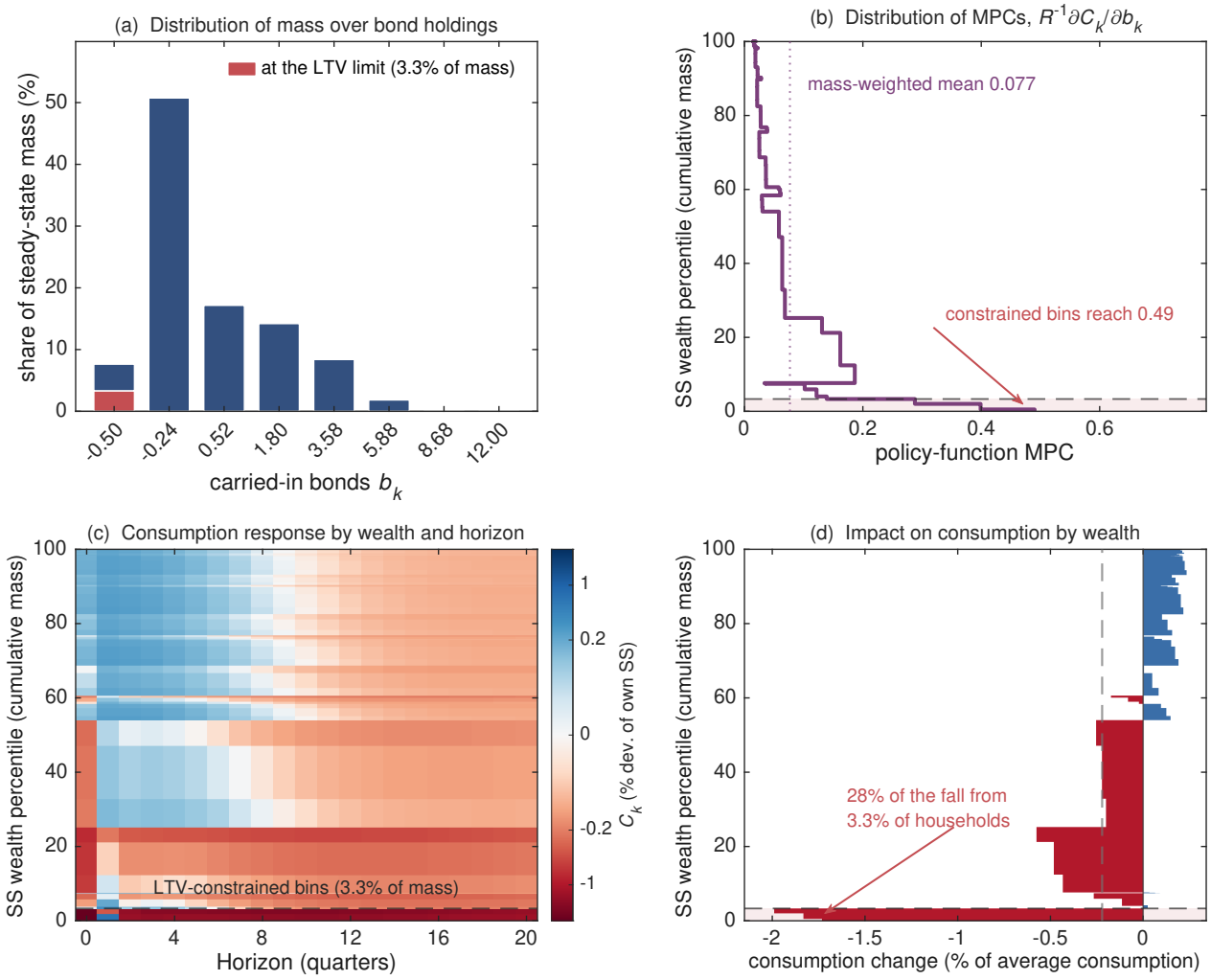
Welfare. Table 5 presents consumption-equivalent welfare under a disagreement shock for the benchmark model and across (1) high/low interest-rate states (i.e., below or above the steady-state interest rate) and (2) recession/expansion (i.e., below and above steady-state output). Table 5 also breaks results down by income quintile and reports the gap between the top and bottom quintiles. Consistent with Figure 6, Table 5 shows that losses are concentrated among poorer households. In addition, the poor-rich gap is wider in a high-rate environment, and recessions hit marginal borrowers (Q1) particularly hard, despite a milder aggregate consumption contraction.

[Table 5 here]

Aggregate responses versus distributional effects. In Figure 3, I plot aggregate responses to a disagreement shock. These responses (e.g., consumption and output) are either small in magnitude or short-lived. To see why, Table 6 decomposes the consumption response by channel and group, alongside the aggregate response. Unconstrained borrowers drive most of the aggregate response, with most of their consumption decline attributable to lower labor income and, to a lesser extent, tighter credit limits. By contrast, the house-price and credit-limit channels are much stronger for constrained borrowers than for their unconstrained counterparts. For constrained borrowers, the credit-limit channel contributes -0.40, labor income contributes -1.28, and house prices contribute -1.42. These numbers suggest that credit tightening also triggers general-equilibrium declines in wages and house prices. In stark contrast, savers experience little to no response to belief-disagreement shocks.

[Table 6 here]

Figure 7: Mass, marginal propensities and consumption decline by wealth



6.2 Why are the Effects Concentrated in the Lower Tail?

To take a closer look at the distributional effects in Figure 6, Figure 7 further breaks down the model's steady-state and impulse-response distributions. Specifically, Panel (a) plots the distribution of bond holdings: about 3.3% of households are at the LTV limit, and about half of the mass is at $b_k = -0.24$. Panel (b) plots the distribution of marginal propensities to consume (MPCs), $R^{-1}\partial C_k/\partial b_k$, computed as finite differences along the bond grid at fixed income and belief states over the steady-state wealth distribution. Panel (c) plots a heatmap of the consumption response by wealth and horizon, and Panel (d) plots the percentage change in consumption by wealth.

Figure 7 shows that only about 3% of households are at the LTV limit (Panel (a)). More importantly, because these households have the highest MPCs (see the bottom left of Panel (b)), their consumption

response to the disagreement shock is the most negative. As shown by the heatmap in Panel (c) and the distribution of consumption responses in Panel (d), constrained borrowers experience the largest consumption declines following the shock. Intuitively, households at the limit cannot borrow to smooth out a shock to income or credit, and thus, they tend to have higher propensities to consume. As Panel (b) shows, low-wealth households are more likely to be close to the constraint and thus tend to have higher MPCs. These households therefore exhibit the largest consumption declines (around 2 percent of average consumption, as in Panel (d)) and account for 28% of the aggregate fall on impact.

6.2.1 Does every contractionary shock fall on the constrained?

So far, I have documented that credit tightening following a belief disagreement shock tends to disproportionately affect marginal borrowers. One natural question is whether more standard shocks could replicate the distributional effects of the disagreement shock.

To that end, I consider the model's responses under (1) a one-standard-deviation shock $v_0 = 1$ for the disagreement shock in Eqn. (10), (2) a monetary tightening shock $\varepsilon_0^R = 0.64\sigma_R$ as in the Taylor rule (22), which translates to 8 basis points per quarter, (3) a TFP shock $\varepsilon_0^A = -0.95$ per Eqn. (17), which decreases A_t by 0.29 percent on impact.

[Table 7 here]

Panel A of Table 7 shows that the effects of monetary tightening tend to spread out across the poorest and richest quintiles, with the cumulative consumption responses of approximately -1%. Turning to TFP shocks, I find that the negative effects are strongest among the poorest households, amounting to around twice the cumulative effects on the richest (Q5) households. Notably, while the aggregate effects are similar in magnitude across disagreement and TFP shocks, it is the disagreement shock that drives the inequality in the consumption response, leading to around 20 times the difference between the poorest and the richest households, compared to around 2 times the difference reported with a TFP shock.

Panel B of Table 7 decomposes the forecast-error variance of consumption across the model's three shocks. It shows that disagreement shocks account for roughly half of the ten-year forecast-error variance of aggregate consumption. This number increases to around 90% for constrained households and around 84% for the poorest households. This result highlights the importance of disagreement shocks in explaining the magnitude of the consumption response, particularly the dynamics of the distributional

effects across households. Table Appendix.4 in the Appendix shows the full variance decomposition with an additional horizon of 4 quarters.

6.2.2 *Income-risk and Average Pessimism Shocks*

Here I consider the model responses to an idiosyncratic income-risk shock. Specifically, the standard deviation of income innovations, σ_e in Section 3.1, becomes a state σ_{et}. It follows Eqn. (10), with σ_{et} in place of σ_{mt}, the calibrated $\bar{\sigma}_e = 0.08$ (Table 3) in place of $\bar{\sigma}_m$, and its own innovation ν_t^e. In response to a one-standard-deviation shock ($\nu_1^e = 1$), consumption responds (cumulatively) about half as much as to a disagreement shock in aggregate (Table 8). The effects of an income-risk shock are also much less pronounced than under disagreement shocks. Intuitively, the shock enters the households' expectations through the transitional probability and moves the income distribution itself. As a result, the shock only works through the standard precautionary saving motives, and not through the credit channel via the loan-to-value limit.

[Table 8 here]

I next consider how the model responds to changes in the *average* belief by adding a shock to the center of belief $\bar{z}$. This shifts the lender's valuation and moves the households across the belief grid. Specifically, $\bar{z}$ in Eqns. (9) and (15) now becomes a state,

$$\bar{z}_t = (1 - \rho_{sm})\bar{z} + \rho_{sm}\bar{z}_{t-1} + \varepsilon_t^{\bar{z}}, \tag{27}$$

where $\bar{z} = \text{logit}\,\bar{m} = 1.10$. I calibrate the shock to tighten the LTV limit by the same amount as the disagreement shock on impact. Overall, while its aggregate responses are similar in magnitude to those of the disagreement shock, the *average* belief shock generates much less pronounced heterogeneity across agents.

6.2.3 *Income vs. Beliefs: Who is most affected?*

Table 9 further breaks down the responses conditional on (1) beliefs and (2) income. Specifically, I consider the cumulative and on-impact responses of consumption and housing to the same disagreement shock, conditional on various income and belief states. Individual beliefs appear to explain less of the overall heterogeneity than income. Intuitively, this dovetails with the lender's inability to sort borrowers

by their beliefs: what matters is the distribution of beliefs, not necessarily who in the distribution holds which belief.

[Table 9 here]

7 Conclusion

This paper asks whether and how disagreement among homebuyers about future house prices generates heterogeneous effects through the credit channel. I document three motivating facts. First, belief disagreement is associated with credit tightening. Second, much of this positive association reflects pessimistic homebuyers. Third, the higher mortgage denial rates associated with disagreement tend to fall on individuals at the lower end of the income distribution.

I then develop a heterogeneous-agent model in which occasionally constrained homebuyers hold heterogeneous beliefs about home prices. I introduce stochastic volatility in the distribution *across* these beliefs, rather than in the distribution of an individual's beliefs over time. A financial intermediary takes the belief distribution into account when choosing how *tight* to set a homebuyer's collateral constraint. I discipline the model using survey and aggregate data, and I find that it matches the data reasonably well. The model predicts that belief disagreement is costly, perhaps especially for agents at the lower end of the wealth and income distribution.

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Table 1: Lending Standards and Belief Disagreements

	$\text{corr}(X_{t-1}, S_t)$		β_1, full		β_1, ex-2020	
Right-hand side X	r	t	β_1	t	β_1	t
Aggregate disagreement						
CRE (avg. of breakouts)	$+\mathbf{0.42}$	3.03	$+9.54$	1.54	$+7.49$	1.17
Residential (GSE-eligible)	$+\mathbf{0.43}$	2.92	$+13.35$	1.18	$+0.87$	0.31
The two halves of the same spread						
Downside half, $p_{50} - p_{10}$	$+\mathbf{0.54}$	4.24	$+1.34$	0.25	$+7.06$	1.78
Upside half, $p_{90} - p_{50}$	$+0.12$	0.80	-6.38	-1.44	-4.59	-1.60
Placebo and near neighbors						
Stock-price disagreement	-0.06	-0.38	$+\mathbf{7.96}$	2.08	$+1.04$	0.48
SCE housing uncertainty	$+0.20$	1.33	$+6.99$	0.65	$+15.23$	1.78
Mean home-price forecast	$-\mathbf{0.75}$	-7.44	$+1.11$	0.25	$+8.06$	1.77
Whose disagreement, D^g on subgroup g only						
Renters	$+\mathbf{0.40}$	2.84	$+13.87$	1.74	$+7.82$	1.23
Under 45	$+\mathbf{0.45}$	3.26	$+10.82$	1.87	$+8.12$	1.43
Young owners	$+\mathbf{0.49}$	3.64	$+4.60$	1.09	$+3.18$	0.76
Owners	$+\mathbf{0.40}$	2.90	$+5.16$	1.06	$+5.57$	1.12
45 and over	$+\mathbf{0.40}$	2.88	$+6.60$	1.13	$+4.91$	0.84

Notes. The left-hand side S_t is SLOOS net percentage tightening, commercial real estate in every row but the second, which swaps in the GSE-eligible residential breakout. Columns 1 and 2 run on $n = 45$ (40 for residential). Columns 3 to 6 spend two quarters on lags and run on $n = 44$ (39 for residential, 40 ex-2020). Bold marks $|t| \geq 1.96$. Each subgroup series is standardized on its own sample.

Table 2: Housing belief disagreement and commercial mortgage leverage, 2016Q4–2024Q4

Specification	LTV change (pp)	NW(4) s.e.	Loans
Disagreement alone	−1.686	0.237	10,956
Add subjective uncertainty	−2.029	0.659	10,956
Add mean expectation	−1.672	0.213	10,956
Add mean and uncertainty jointly	−0.838	2.179	10,956
Add lagged LTV	−1.205	0.400	10,514
Quarterly first differences	−0.009	0.650	10,514
Four-quarter differences	−1.772	0.339	9,110
Michigan disagreement	−1.486	0.259	10,956

Notes. Each row reports the coefficient on lagged national disagreement from a separate quarterly OLS regression with an intercept. The outcome is mean securitization LTV in percentage points, adjusted for property type, state, and log appraised value. LTV is scheduled principal at securitization divided by the corresponding appraisal. Loans are grouped by origination quarter. Source: Form ABS-EE disclosures accessed through WRDS and reconciled with SEC filings. The full sample contains 10,956 whole loans over 2016Q4–2024Q4.

Table 3: Calibration

Parameter	Description	Value	Source/Target
Household preferences and constraints			
β_h	Discount factor	0.99	patience rate $\approx 1\%$/qtr. $r^\star$ cleared endogenously
j	Housing preference weight	0.10	chosen. The weight in Iacoviello (2005)
σ_c	CRRA, consumption	2	Chetty (2006)
σ_h	CRRA, housing	3	-
ξ	Inverse Frisch elasticity	2	Frisch elasticity $= 0.5$, Chetty (2012)
$\bar{\phi}$	Household LTV limit	0.80	mortgage-market convention
ψ	Labor disutility weight	1.07	endogenous: $\bar{\psi} = \bar{w}\bar{\lambda}$, normalizing $\bar{l} = 1$
Idiosyncratic productivity			
ρ_e, σ_e	Persistence, innovation s.d.	0.95, 0.08	quarterly HA range; σ_e set for bond-market clearing
N_e	Productivity states	3	Rouwenhorst (1995); Kopecky and Suen (2010)
Entrepreneur and production			
β_e	Discount factor	0.98	Iacoviello (2005)
σ_{ce}	CRRA, entrepreneur	2	-
α	Real-estate share	0.33	-
Ξ_e	Entrepreneur LTV limit	0.10	-
Nominal rigidity, housing, policy			
γ, θ	Calvo, elasticity of subst.	0.75, 5	4-quarter price duration; 25% markup
γ_w	Wage partial adjustment	0.90	Blanchard and Gali (2007)
$\chi_h, \chi_{\Delta I}$	Housing adjustment costs	10, 3.5	chosen; form from Topel and Rosen (1988), Christiano et al. (2005)
δ_h	Housing depreciation	0.025	NIPA-consistent (quarterly)
π	Inflation target	$1.02^{1/4}$	2% annual
Transmission and additional mechanisms			
χ_q	Lender's quantile multiplier	8.59	Estimated from SCE and SLOOS data (Table 1)
ζ_A	Endogenous TFP feedback	0.05	-
Belief process and remaining shocks			
$\bar{m}$	Steady-state cognitive discount	0.75	Meggiorini (2023) and Vu (2026)
ρ_m	Persistence of m_{it}	0.92	Vu (2026)
$\bar{\sigma}_m$	Steady-state σ_{mt}	0.05	-
ρ_{sm}	Persistence of $\log \sigma_{mt}$	0.942	-
σ_v	S.d. of disagreement innovation	0.066	-
ρ_A, σ_A	TFP process	0.90, 0.003	-
ρ_R, ϕ_Y, ϕ_π	Taylor rule	0.84, 0.12, 1.50	Taylor principle
σ_R	Monetary policy shock	0.00125	12.5 bp quarterly (50 bp annualized)

Table 4: Aggregate Moments: Model vs. Data

Moment	Data	Model
Panel A. Volatility. *Standard deviation of the cyclical component, in percent*		
Output	1.04	1.30 [1.15, 1.48]
Consumption	0.65	0.45 [0.38, 0.52]
Real wage	1.05	0.87 [0.69, 1.09]
Real house price	3.02	1.72 [1.52, 1.96]
Real-estate investment	4.94	6.14 [5.36, 7.08]
the same, relative to output	4.75	4.70 [4.47, 4.98]
Real-estate credit	2.41	1.71 [1.29, 2.39]
Policy rate (annualized pp)	1.07	0.83 [0.66, 1.09]
Panel B. Comovement. *Contemporaneous correlation*		
House price with consumption	0.55	0.49 [0.29, 0.64]
House price with real-estate credit	0.25	0.30 [0.12, 0.50]
Consumption with real-estate credit	0.30	0.31 [0.03, 0.56]
Consumption with real-estate investment	0.56	0.43 [0.21, 0.61]
Real-estate credit with the policy rate	0.37	0.22 [−0.02, 0.41]
Inflation with the policy rate	0.36	0.32 [0.15, 0.46]
Output with consumption	0.83	0.63 [0.47, 0.74]
Panel C. Persistence. *Autocorrelation of the cyclical component*		
Real-estate credit, one quarter	0.90	0.76 [0.55, 0.87]
two quarters	0.76	0.64 [0.42, 0.77]
four quarters	0.45	0.38 [0.18, 0.55]
Policy rate, one quarter	0.93	0.81 [0.72, 0.87]
Real wage, two quarters	0.47	0.55 [0.41, 0.67]
Panel D. Lead and lag. $\mathrm{corr}(x_{t-k}, y_t)$*: the first series leads by* k		
Real-estate investment leads the policy rate, $k = 2$	0.52	0.47 [0.30, 0.60]
Policy rate leads real-estate credit, $k = 1$	0.42	0.31 [0.06, 0.50]
Real-estate credit leads real-estate investment, $k = 4$	−0.41	−0.37 [−0.51, −0.20]
Real-estate credit leads mortgage debt, $k = 4$	0.40	0.38 [0.14, 0.59]
Output leads inflation, $k = 2$	0.19	0.22 [0.04, 0.38]
Panel E. The distribution		
House price with the bottom-50% net worth share	0.65	0.82 [0.66, 0.90]
Consumption with that share	0.33	0.25 [−0.02, 0.48]
Output leads that share, $k = 1$	0.39	0.39 [0.20, 0.55]

Notes. Data are quarterly, 1987Q1–2019Q4 ($n = 132$). The three wealth-share rows begin in 1989Q3. Real-estate investment is residential plus nonresidential structures; real-estate credit is loans secured by real estate at all commercial banks, compared with household mortgage debt plus entrepreneur borrowing in the model. I apply an HP filter with $\lambda = 1600$ to $100\times$ logs, so standard deviations are in percent; inflation, the policy rate, and the wealth share are in percentage points and are filtered in levels. The model column reports the median across 1,000 simulated 132-quarter samples, and brackets report the 5–95% range across simulations.

Table 5: Welfare across initial states and wealth groups

Initial state	Overall	Q1	Q2	Q3	Q4	Q5	Q1−Q5
A. Impact consumption equivalent (%)							
Steady-state benchmark	1.2116	-0.1404	0.8308	1.3986	1.7527	2.0030	-2.1434
Pooled accepted states	1.2253	-0.1626	0.8066	1.4184	1.8037	2.0492	-2.2119
Low interest rate	1.2461	-0.1396	0.8177	1.4349	1.8317	2.0802	-2.2198
High interest rate	1.1742	-0.2225	0.7776	1.3787	1.7370	1.9750	-2.1974
Recession	1.2200	-0.1563	0.8043	1.4124	1.7921	2.0386	-2.1949
Expansion	1.2373	-0.1505	0.8220	1.4277	1.8144	2.0621	-2.2126
B. Long-horizon annuitized welfare equivalent (%)							
Steady-state benchmark	-0.0644	-0.2629	-0.0666	-0.0248	-0.0165	-0.0087	-0.2542
Pooled accepted states	-0.0580	-0.2241	-0.0605	-0.0250	-0.0171	-0.0108	-0.2133
Low interest rate	-0.0578	-0.2211	-0.0600	-0.0256	-0.0175	-0.0115	-0.2096
High interest rate	-0.0579	-0.2282	-0.0611	-0.0237	-0.0162	-0.0091	-0.2191
Recession	-0.0598	-0.2351	-0.0623	-0.0249	-0.0169	-0.0102	-0.2249
Expansion	-0.0578	-0.2221	-0.0603	-0.0253	-0.0173	-0.0110	-0.2111

Units. Both panels are expressed in consumption-equivalent percentages. B uses 1,600 quarters; $\beta_h = 0.99$. Negative values indicate losses. Utility includes consumption, housing services, and labor disutility. Each state has 5,000 draws.

Table 6: Impact Responses by Groups

	Impact consumption response (% of group consumption)			
		Borrowers		
Channel	Aggregate	Constrained	Unconstrained	Savers
Credit limit ($\tilde{\phi}^{\text{eff}}$)	-0.10	-0.40	-0.13	-0.04
Real rate (R, π)	$+0.16$	-0.18	$+0.10$	$+0.25$
Labor income (w, L_e)	-0.31	-1.28	-0.43	-0.13
House price (q)	-0.05	-1.42	-0.05	$+0.00$
Transfers (F, Π_c)	$+0.14$	$+0.47$	$+0.21$	$+0.05$
Belief transition (σ_m)	$+0.00$	-0.00	$+0.00$	$+0.00$
All channels	-0.16	-2.80	-0.30	$+0.12$

Notes. Impact consumption response by channel and group, in percent of group consumption. The three groups partition the bins by the sign of steady-state carried-in bonds b_k and by whether the collateral constraint binds. *Constrained* borrowers are the three bins at the loan-to-value limit, all at $b_k = -0.50$ (3.3% of mass). *Unconstrained* borrowers are the remaining bins with $b_k < 0$ (55.1%). *Savers* are the bins with $b_k > 0$ (41.6%), where k denotes the bin.

Table 7: Disagreement versus Monetary and TFP Shocks

	Disagreement	Monetary	TFP
Panel A. Consumption responses to shocks of equal aggregate size			
Shock size (standard deviations)	1.00	0.64	0.95
Aggregate consumption, impact	−0.16	−0.16	−0.16
Aggregate consumption, cumulative	−3.82	−0.95	−3.56
Constrained households, impact	−2.85	−0.98	+0.22
Constrained households, cumulative	−48.30	−1.53	−8.50
Constrained share of the impact decline (%)	28.2	13.7	–
Poorest quintile (Q1), cumulative	−18.51	−1.08	−5.43
Richest quintile (Q5), cumulative	−0.78	−0.86	−2.73
Panel B. Share of the ten-year forecast-error variance of consumption (%)			
All households	45.9	8.8	45.4
Constrained households	89.6	4.6	5.8
Poorest quintile (Q1)	83.6	3.8	12.6
Richest quintile (Q5)	50.3	8.4	41.3
Gap: Q1 minus Q5	92.9	2.6	4.6
Gap: constrained minus all	92.5	4.2	3.3

Notes. Panel A reports consumption responses in percent of each group's steady-state consumption. Cumulative responses are undiscounted sums over $h = 0, \ldots, 40$ quarters. Each shock is scaled so that aggregate consumption falls by the same 0.16 percent on impact as after a one-standard-deviation disagreement shock. The monetary shock is a tightening, a positive innovation ε_t^R (one standard deviation is 12.5 basis points per quarter); the TFP shock is a contraction, the negative of the response to a positive innovation ε_t^A.

Table 8: Disagreement versus Income-Risk and Average-Pessimism Shocks

	Disagreement	Income risk	Average pessimism
Shocked variable	σ_{mt}	σ_{et}	$\bar{z}_t$
Panel A. Loan-to-value limit			
Impact	−0.59	0.00	−0.59
Largest tightening (quarter)	−2.16 (8)	–	−0.59 (0)
Cumulative	−55.33	0.00	−9.25
Panel B. Aggregate consumption			
Impact	−0.16	−0.28	−0.06
Cumulative	−3.54	−1.89	−0.66
Panel C. Consumption by group			
Constrained households, impact	−2.85	−1.27	−1.61
Constrained households, cumulative	−48.30	+2.87	−7.89
Constrained share of the impact decline (%)	28.2	8.9	44.3
Constrained share of the cumulative decline (%)	27.7	–	28.2
Poorest quintile (Q1), cumulative	−18.51	−1.24	−2.85
Richest quintile (Q5), cumulative	−0.78	−2.64	−0.23
Low-income pessimists, cumulative	−6.46	+4.56	−0.99
Low-income optimists, cumulative	−19.31	+5.97	−2.88

Notes. Responses in percent. The disagreement shock is a one-standard-deviation innovation, $\nu_0 = 1$ in Eqn. (10). The income-risk shock is an innovation of the same size and persistence to the standard deviation of income innovations, σ_{et}, which rises from 0.080 to 0.085 on impact; labor income is normalized so that aggregate labor income does not change. The average-pessimism shock is an innovation $\varepsilon_0^{\bar{z}} = -0.0118$ in Eqn. (27), sized to tighten the loan-to-value limit on impact by as much as the disagreement shock. Cumulative responses are undiscounted sums over $h = 0, \ldots, 40$ quarters. Panel A reports the loan-to-value limit $\bar{\phi}_t^{\text{eff}}$ of Eqn. (16); the largest tightening is its lowest value over these quarters, with the quarter in parentheses. Panel B reports the model's aggregate consumption, as in Figure 3, which includes the movement of households across the wealth distribution. Panel C reports consumption in percent of each group's steady-state consumption, built as in Table 7. All responses use the second-order solution. Quintiles are the steady-state wealth quintiles of Figure 6.

Table 9: Who Bears the Disagreement Shock: Income and Beliefs

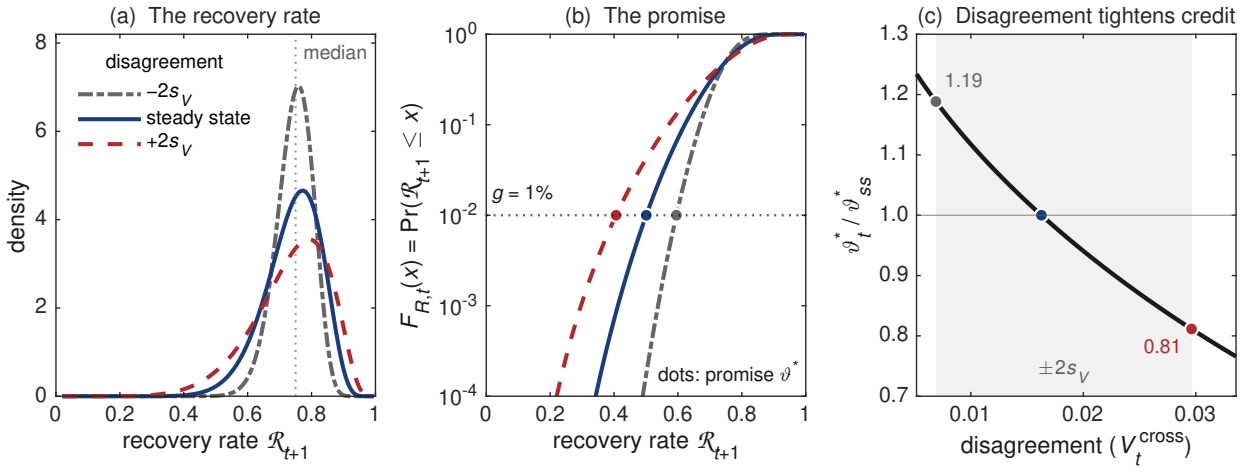
			At the limit	Consumption		Housing
Income	Belief	Mass (%)	(% of cell)	Impact	Cumulative	Cumulative
Low	Pessimist ($m = 0.712$)	6.3	8.0	−0.58	−6.5	−2.7
	Median ($m = 0.750$)	12.3	12.2	−0.72	−10.5	−5.3
	Optimist ($m = 0.784$)	6.3	20.7	−0.99	−19.3	−11.0
Middle	Pessimist	12.7	0.0	−0.07	−1.5	+1.0
	Median	24.6	0.0	−0.10	−2.3	+0.7
	Optimist	12.7	0.0	−0.15	−3.8	+0.1
High	Pessimist	6.3	0.0	+0.17	−0.6	+1.6
	Median	12.3	0.0	+0.15	−0.8	+1.8
	Optimist	6.3	0.0	+0.14	−1.0	+2.0

Notes. Responses to a one-standard-deviation disagreement shock, in percent of each cell's steady-state level, built as in Figure Appendix.3: the cell aggregate is weighted by steady-state mass. Cumulative responses are undiscounted sums over $h = 0, \ldots, 40$ quarters, as in Table 7. Income cells are the model's three income states ($e = 0.674, 0.968, 1.391$); belief cells are its three belief states. "At the limit" is the share of the cell's steady-state mass at the loan-to-value limit.

Appendix to "Belief Disagreement and the Housing Credit Channel"

A The Lender's Underwriting Problem

Figure Appendix.1: Recovery risk, the optimal promise, and the credit limit



Notes. Panel (a): densities of $\mathcal{R}_{t+1}$, where logit $\mathcal{R}_{t+1} = \bar{z} + \kappa_R \sqrt{V_t^{\text{cross}}}\, \varepsilon_{t+1}$, Eqn. (14). *Panel (b)*: distribution functions, with the promise at $F_{R,t}(\vartheta_t^*) = g$. *Panel (c)*: $\vartheta_t^* / \vartheta_{ss}^*$ against V_t^{cross}, with logit $\vartheta_t^* = \bar{z} - \chi_q \sqrt{V_t^{\text{cross}}}$. It scales both limits, $\bar{\phi}_t^{\text{eff}} = \bar{\phi}\, \vartheta_t^* / \vartheta_{ss}^*$ and $\Xi_{e,t}^{\text{eff}} = \Xi_e\, \vartheta_t^* / \vartheta_{ss}^*$, Eqns. (16) and (19).

Here, I derive Eqn. (15) allowing the loss budget to bind and provide details on the comparative statics of χ_q. In levels, the lender advances $\mathcal{B}_t$ at the gross loan rate R_t^L, funds it at $R_t^f < R_t^L$, receives $\min\{R_t^L \mathcal{B}_t, \mathcal{R}_{t+1} \mathcal{C}_t\}$ at $t+1$, and keeps expected credit losses within a fixed limit $\bar{\ell}$ per unit of collateral:

$$\max_{0 \le R_t^L \mathcal{B}_t \le \mathcal{C}_t} \mathbb{E}_t^{\text{lend}} \left[\min\{R_t^L \mathcal{B}_t, \mathcal{R}_{t+1} \mathcal{C}_t\} \right] - R_t^f \mathcal{B}_t \quad \text{s.t.} \quad \mathbb{E}_t^{\text{lend}} \left[\max\{R_t^L \mathcal{B}_t - \mathcal{R}_{t+1} \mathcal{C}_t, 0\} \right] \le \bar{\ell}\, \mathcal{C}_t. \tag{28}$$

Dividing by $\mathcal{C}_t$, writing $\vartheta = R_t^L \mathcal{B}_t / \mathcal{C}_t$ and $g_t = 1 - R_t^f / R_t^L$, and using $\min\{\vartheta, \mathcal{R}\} = \vartheta - \max\{\vartheta - \mathcal{R}, 0\}$ gives Eqn. (12) with the loss limit attached. For a recovery distribution with a continuous positive density on $(0,1)$, the objective is strictly concave, and the Kuhn–Tucker conditions with multiplier $\eta_t \ge 0$ on the limit yield

$$F_{R,t}(\vartheta_t^*) = \frac{g_t}{1 + \eta_t}. \tag{29}$$

The optimal promise is therefore a quantile of the recovery distribution. The lender tolerates a shortfall probability equal to its margin, discounted by the shadow price of loss capacity.

Under Eqn. (14), $F_{R,t}(x) = \Phi\big((\operatorname{logit} x - \bar{z})/(\kappa_R \sqrt{V_t^{\text{cross}}})\big)$, and inverting Eqn. (29) gives

$$\operatorname{logit} \vartheta_t^* = \bar{z} - \kappa_R \, \Phi^{-1}\Big(1 - \frac{g_t}{1+\eta_t}\Big) \sqrt{V_t^{\text{cross}}}, \tag{30}$$

which is Eqn. (15) when the limit is slack ($\eta_t = 0$) and the margin is constant. Figure Appendix.1 plots Eqn. (30) at the calibrated values. As disagreement rises, the recovery distribution stretches about a fixed median, and the promise moves left along the horizontal line at g so that the shortfall probability is unchanged. Panel (c) plots the factor $\vartheta_t^*/\vartheta_{ss}^*$ that scales both loan-to-value limits, which falls as disagreement rises.

B Numerical Computation

This appendix records the numerical choices behind Section 4.

Grids. Productivity e_{it} is discretized using the Rouwenhorst method and normalized so that $\mathbb{E}[e] = 1$. The cognitive discount factor is discretized in the logit space z_{it} of Eqn. (9) using the Tauchen method, which keeps m_{it} inside $(0, 1)$ at every node.

Steady state. The household problem is solved by Coleman iteration of the consumption policy to a tolerance of 10^{-9}, with damping.

Perturbation and checks. The model is perturbed at second order with pruning, with simulated paths built from the recursion in Andreasen et al. (2018). The Blanchard–Kahn conditions hold, and bond market clearing is the imposed condition with goods market clearing left as the redundant one. Figures Appendix.6 and Appendix.7 repeat the exercise at third order on the same grid.

C Survey Data

This appendix presents additional details on the survey data used in the main text.

Survey of Consumer Expectations. `Q31v2` asks about the direction of the expected change in the nationwide home price over the next twelve months and `Q31v2part2` elicits the magnitude. The signed

point forecast is winsorized at the pooled first and ninety-ninth percentiles and aggregated to the quarter without weights. The sample consists of 179,946 person-waves over the 49 quarters from 2013Q2 to 2025Q2. Disagreement D_t is the within-quarter standard deviation of that forecast. The uncertainty series in Figure 1 is instead the mean across respondents of each respondent's own density interquartile range from item `C1`, so it is a first moment of individual second moments. The stock-price placebo is built the same way from `Q6new`, the reported percent chance that the market is higher in twelve months.

Panels cut from the SCE. The state panel uses the state a respondent reports in that wave, which gives 2,475 state-quarter cells across 52 state codes with a median of 49 respondents per cell. The subgroup rows of Table 1 recompute D_t within age, tenure, and the intersection of the two, where the young are under 45 and the borrower pool is young owners.

Senior Loan Officer Opinion Survey (SLOOS). Credit tightening S_t is the average of the three commercial-real-estate breakouts named in Section 2.2, which the survey reports from 2013Q4 onward. Where a longer window is needed, the series is spliced at that quarter to *DRTSCREL*, the single commercial-real-estate question that preceded the breakouts. The residential row of Table 1 uses *SUBLPDHMSENQ*, the GSE-eligible mortgage breakout, which begins in 2015Q1. The estimation sample is the 45 quarters from 2014Q1 to 2025Q1.

Michigan Survey. Figure Appendix.5, the credit correlations in Figure Appendix.2, and Table Appendix.3 all use the cross-sectional standard deviation of *HOMPX1*, the signed expected twelve-month percent change in the prices of homes like the respondent's.

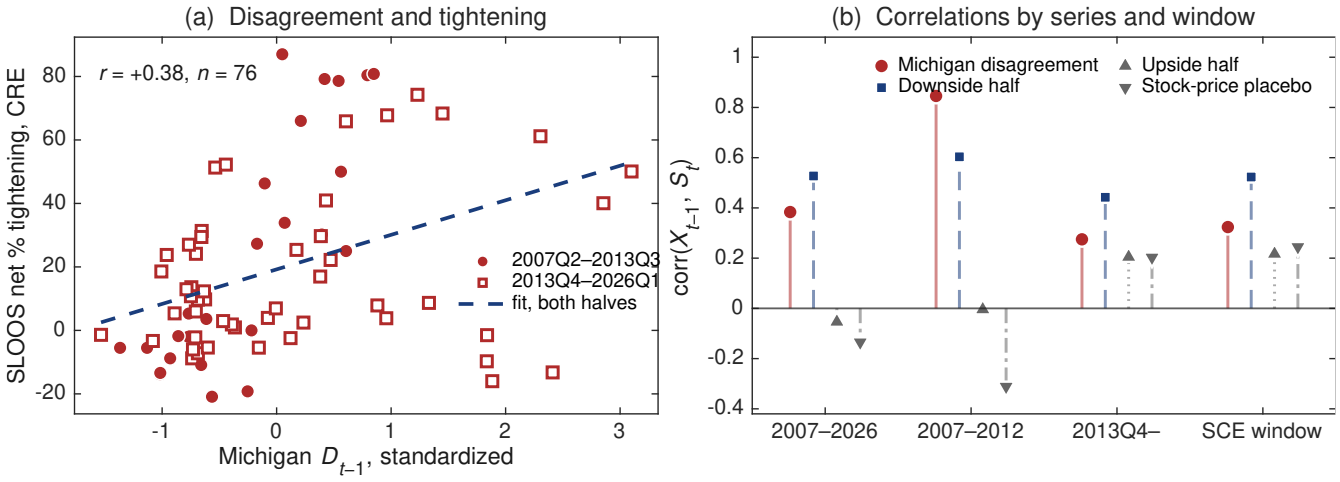
D Alternative Evidence from the Michigan Survey

I check whether the motivating facts are also present in the Michigan Survey of Consumers.[14] Panel (a) of Figure Appendix.2 plots the correlation between credit tightening (series *DRTSCREL* before 2013Q4 and the average of *SUBLPDRCSC*, *SUBLPDRCSN*, and *SUBLPDRCSM* as in Figure 2) and the analogous disagreement measure D_{t-1} computed from the Michigan survey. Panel (b) plots the same correlation

[14] Whereas the SCE asks respondents about the **nationwide** home price (i.e., "the average home price **nationwide**"), the Michigan survey asks about local prices (i.e., "the prices of homes like yours in **your community**"). The Michigan series used here is *HOMPX1*, the signed expected percent change in the prices of homes like the respondent's over the next 12 months. Source: `https://data.sca.isr.umich.edu/subset/subset.php`. Retrieved September 5, 2026.

for overall disagreement, decomposed into the upper ($p_{90,t} - p_{50,t}$) and lower ($p_{50,t} - p_{10,t}$) halves, and reports a stock-price placebo across different sample windows. I also test whether the lower/upper-tail asymmetry is specific to housing by constructing analogous measures for expectations about stock prices (*PSTK*).[15]

Figure Appendix.2: Disagreements in the Michigan Survey



Notes. Panel (a) plots SLOOS net percentage tightening on commercial real estate against standardized Michigan disagreement one quarter earlier, 2007Q2–2026Q1. Filled circles are the quarters carried by the discontinued single CRE question (*DRTSCREL*: Net Percentage of Domestic Banks Tightening Standards for Commercial Real Estate Loans) and open squares those carried by the mean of the three series (as in Figure 2), and the fit is on both. Panel (b) plots the same simple correlation between disagreement D_{t-1} and tightening S_t, along with a stock-price placebo.

Figure Appendix.2 shows that credit tightening is positively correlated with disagreement in the Michigan survey. In addition, the correlation appears to be driven by respondents who predict low house-price growth (i.e., the downside half of the belief distribution).

[15] In the Michigan survey, *PSTK* reports the "Probability of Increase in Stock Market in Next Year." Source: `https://data.sca.isr.umich.edu/subset/subset.php`. Retrieved September 5, 2026.

Table Appendix.1: Related Literature: A Comparison

Paper	Shares with this paper	Does not have
Ludwig et al. (2024)	survey-measured belief heterogeneity in a housing model with borrowing limits	lender pricing of dispersion, dispersion shock
Dong et al. (2022)	belief cut-off at a given loan-to-value ratio, credit supply meets beliefs	dispersion-dependent limit
Tomura (2013)	borrower–lender belief gap in an Iacoviello (2005) economy	belief continuum, quantile rule
Guerreiro (2023)	disagreement amplifies business cycles through aggregate demand with heterogeneous households	housing, lender, credit supply
Dong et al. (2025)	disagreement raises the constrained share	housing collateral, priced limit
Forlati and Lambertini (2011) and Christiano et al. (2014)	dispersion shock tightens credit terms	beliefs, survey measurement
Diamond and Landvoigt (2022)	household leverage priced by intermediaries, heterogeneous (young and middle-aged) borrowers	belief dispersion as the state

Table Appendix.2: The series behind Figure 4 and Table 4

Panel A. The six series behind Figure 4. *2013Q2–2025Q2, unfiltered.*

Fig. 4	Model	Source identifier, and series	Deflator
Left-hand sides. Each enters as $100\Delta \log x_t$*; at horizon* h*, the left-hand side is* $\sum_{s=0}^{h} \Delta \log x_{t+s}$.			
(a) Output	Y_t	`GDPC1`. Real gross domestic product, chained dollars	—
(b) Consumption	C_t	`PCECC96`. Real personal consumption expenditures	—
(c) Real wage	w_t	`ECIWAG`. Employment Cost Index, wages and salaries, private industry	`GDPDEF`
(d) House price	q_t	`CSUSHPISA`. S&P CoreLogic Case–Shiller U.S. national home price index	`GDPDEF`
(e) Housing quantity	I_{ht}	`PRFIC1`. Real private residential fixed investment	—
(f) Credit quantity	$\int B_{it}\, di$	`CREACBM027SBOG`. Loans secured by commercial real estate, all commercial banks	`GDPDEF`
Right-hand side. Standardized over the estimation sample.			
u_t	σ_{mt}	Cross-sectional standard deviation of signed twelve-month home-price point forecasts: FRBNY Survey of Consumer Expectations public Core microdata, item `Q31v2` signed by `Q31v2part2`, winsorized 1/99 pooled, unweighted, and aggregated to the quarter	—

Panel B. The eleven series behind Table 4. *1987Q1–2019Q4; every series HP-filtered at* $\lambda = 1600$.

Variable	Source identifier, and series	Transformation	Variable	Source identifier, and series	Transformation
Output Y_t	`GDPC1`. Real gross domestic product, chained dollars	100 log, HP	Real-estate credit	`REALLN`. Real estate loans, all commercial banks	÷`GDPDEF`, 100 log, HP
Consumption C_t	`DNDGRA3Q086SBEA` + `DSERRA3Q086SBEA`. Real PCE, nondurables and services	Törnqvist (i.e., weighted geometric average), 100 log, HP	Mortgage debt	`HHMSDODNS`. Households, one-to-four-family mortgage debt	÷`GDPDEF`, 100 log, HP
Real wage w_t	`COMPRNFB`. Real hourly compensation, nonfarm business	100 log, HP	Inflation π_t	`GDPDEF`. GDP implicit price deflator	400Δ log, level, HP
Real house price q_t	`CSUSHPISA`. Case–Shiller U.S. national home price index	÷`GDPDEF`, 100 log, HP	Policy rate R_t	`FEDFUNDS`. Effective federal funds rate	level, HP
Real-estate investment I_{ht}	`A011RA3Q086SBEA` + `B009RA3Q086SBEA`. Real fixed investment, residential and nonresidential structures	Törnqvist, 100 log, HP	Bottom-50% net worth share	`WFRBSB50215`. Net worth share of the bottom 50%, DFA	level, HP
Housing stock $\int H_{it} di$	`HNOREMQ027S`. Households, real estate at market value	÷`CSUSHPISA`, 100 log, HP			

Notes. Data are from the St. Louis Fed's FRED database. Monthly series are averaged to quarterly frequency.

Table Appendix.3: Lending Standards and Belief Disagreements: Evidence from the Michigan Survey

	corr(X_{t-1}, S_t)		β_1, full		β_1, ex-2020	
Right-hand side X	r	t	β_1	t	β_1	t
Aggregate disagreement						
Michigan disagreement D^{MSC}	**+0.38**	3.57	+0.24	0.06	+1.54	0.67
mode-adjusted	**+0.37**	3.48	+0.17	0.04	+1.51	0.65
Residential (spliced)	**+0.28**	2.45	+9.56	1.54	−0.81	−0.40
The two halves of the same spread						
Downside half, $p_{50} - p_{10}$	**+0.53**	5.34	−4.73	−1.38	**−6.96**	−2.66
Upside half, $p_{90} - p_{50}$	−0.05	−0.46	+1.16	0.64	+1.26	0.74
Placebo and near neighbors						
Stock-price disagreement (PSTK)	−0.13	−1.17	+0.19	0.14	+0.52	0.38
Mean home-price forecast	**−0.38**	−3.58	+8.48	1.96	**+11.47**	3.12
Whose disagreement						
Owners	**+0.38**	3.57	+0.24	0.06	+1.54	0.67
Young owners	**+0.35**	3.23	+1.66	0.71	+2.93	1.82
45 and over	**+0.39**	3.65	−0.05	−0.02	−0.15	−0.08
Under 45	**+0.35**	3.23	+1.66	0.71	+2.93	1.82

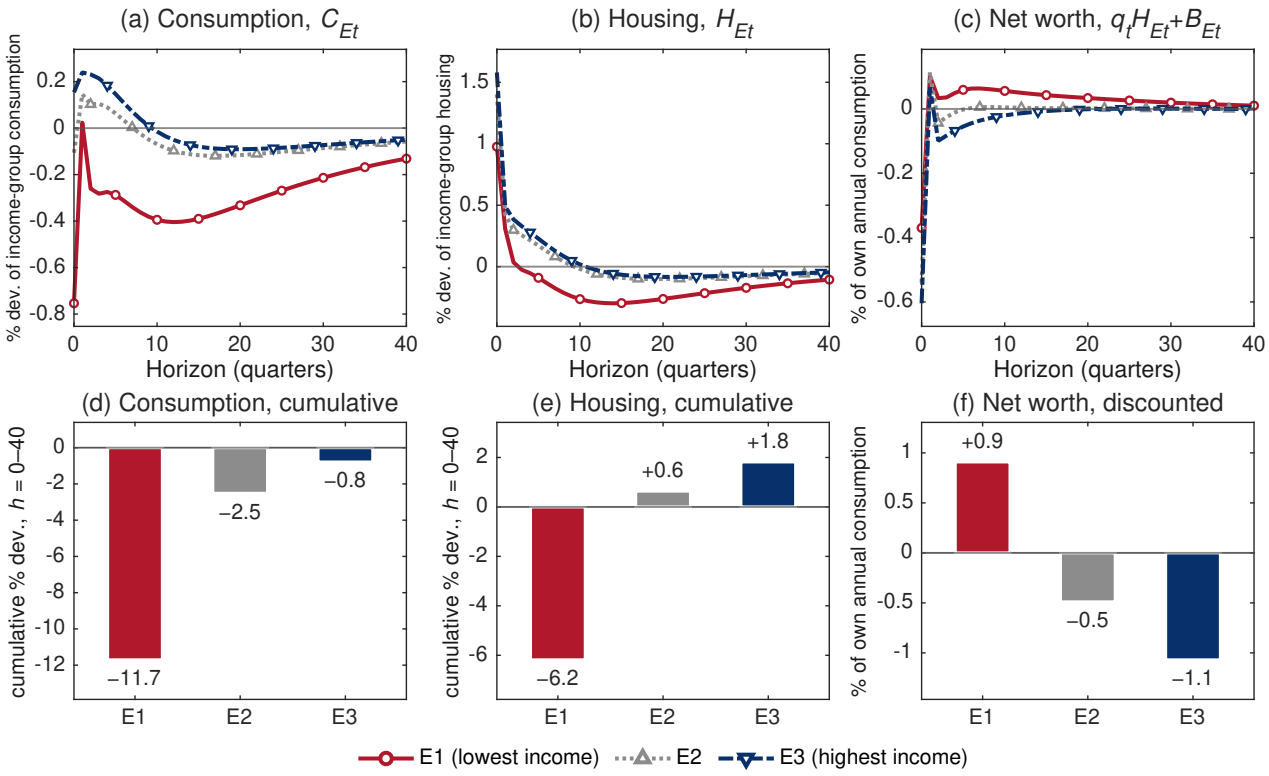
Notes. The layout follows Table 1, and the estimator is the same. S_t is net percentage tightening on commercial real estate in every row but the third, spliced at 2013Q4 from DRTSCREL to the mean of the three breakouts. Every Michigan respondent asked the home-price question is a homeowner, so the Owners row reproduces the full sample and the Young owners row reproduces the Under 45 row.

Table Appendix.4: Forecast-Error Variance Decomposition of Consumption

	One year ($H = 4$)			Ten years ($H = 41$)		
Consumption of	Disagreement	Monetary	TFP	Disagreement	Monetary	TFP
All households	15.6	17.2	67.2	45.9	8.8	45.4
Q1 (poorest)	56.9	14.6	28.6	83.6	3.8	12.6
Q2	32.2	13.5	54.3	51.1	7.4	41.5
Q3	19.0	18.5	62.5	32.1	11.7	56.3
Q4	35.8	9.9	54.3	42.2	8.7	49.1
Q5 (richest)	52.2	8.4	39.4	50.3	8.4	41.3
Constrained households	69.1	18.0	12.9	89.6	4.6	5.8
Gap: Q1 minus Q5	81.5	9.0	9.5	92.9	2.6	4.6
Gap: constrained minus all	75.5	16.2	8.3	92.5	4.2	3.3

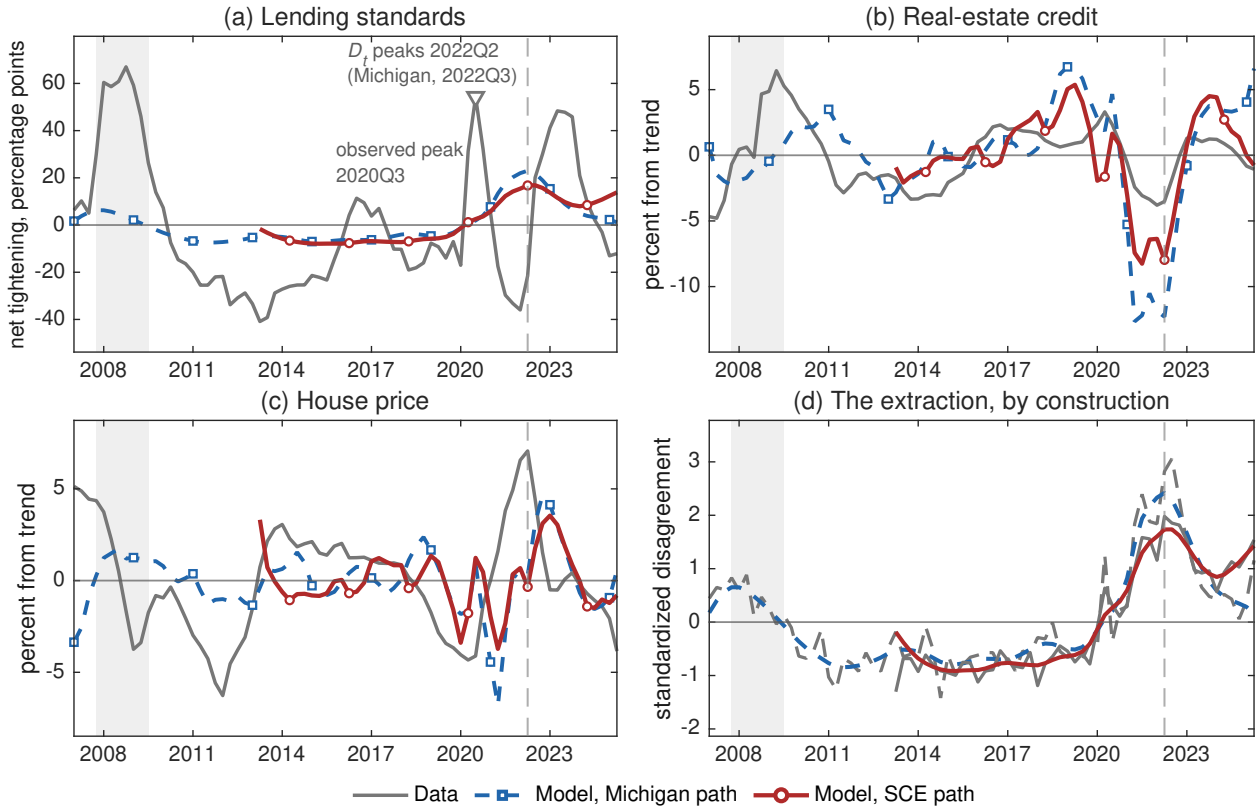
Notes. Share of the H-quarter-ahead forecast-error variance of each group's consumption (in percent of its steady-state level) due to each of the model's three shocks, computed from one-standard-deviation impulse responses of the second-order solution: the share of shock s is $\sum_{h<H} r_{s,h}^2 / \sum_{s'} \sum_{h<H} r_{s',h}^2$. The gaps use the difference between the two groups' responses. Quintiles are the steady-state wealth quintiles of Figure 6. Shares are relative to the model's three shocks. For comparison, in HP-filtered simulations the disagreement shock accounts for 19 percent of the variance of aggregate consumption. Panel B of Table 7 repeats the ten-year shares of selected groups.

Figure Appendix.3: Consumption, housing, and net worth by income



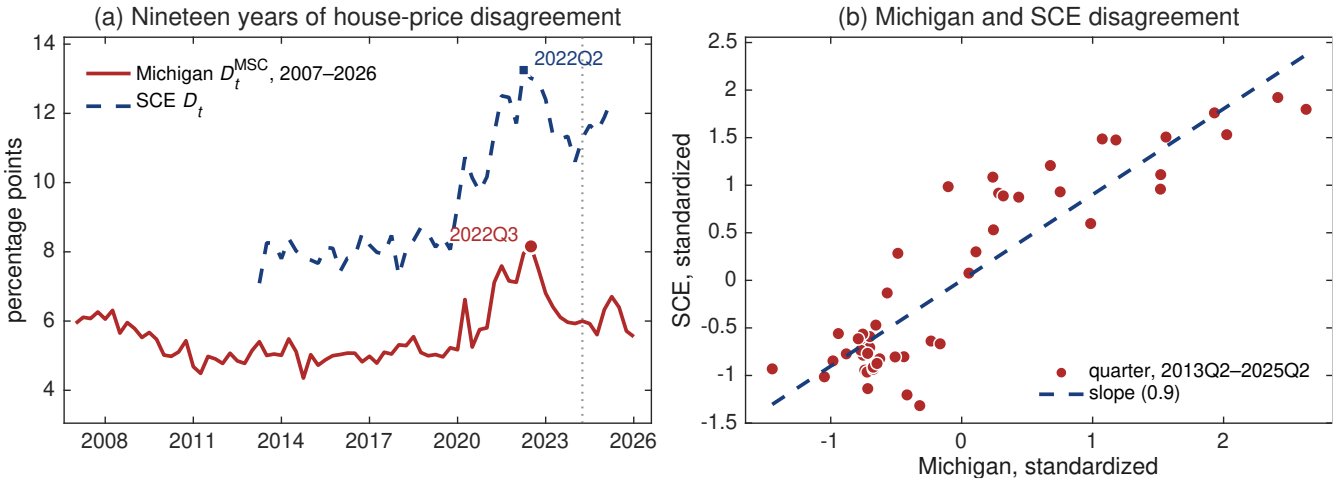
Notes. The six panels of Figure 6, on the same solve and conventions, with the partition changed from wealth quintiles to the model's three income states. Writing E for an income group (as Q denotes a wealth quintile), $C_{Et} = \sum_{k \in E} \omega_{k,ss} C_{kt}$ and likewise for housing and net worth, normalized exactly as in Figure 6: panels (a), (b), (d), and (e) are percent deviations of the group aggregate divided by that group's steady-state total, with (d) and (e) the undiscounted cumulative sums of the series above them; panels (c) and (f) plot $q_t H_{Et} + B_{Et}$ relative to its no-shock path, mass-weighted to a group mean and divided by four times that group's steady-state consumption, with (f) discounted at $\beta = 0.99$ over the full path rather than cumulated to $h = 40$.

Figure Appendix.4: Historical implications of measured housing-belief disagreement



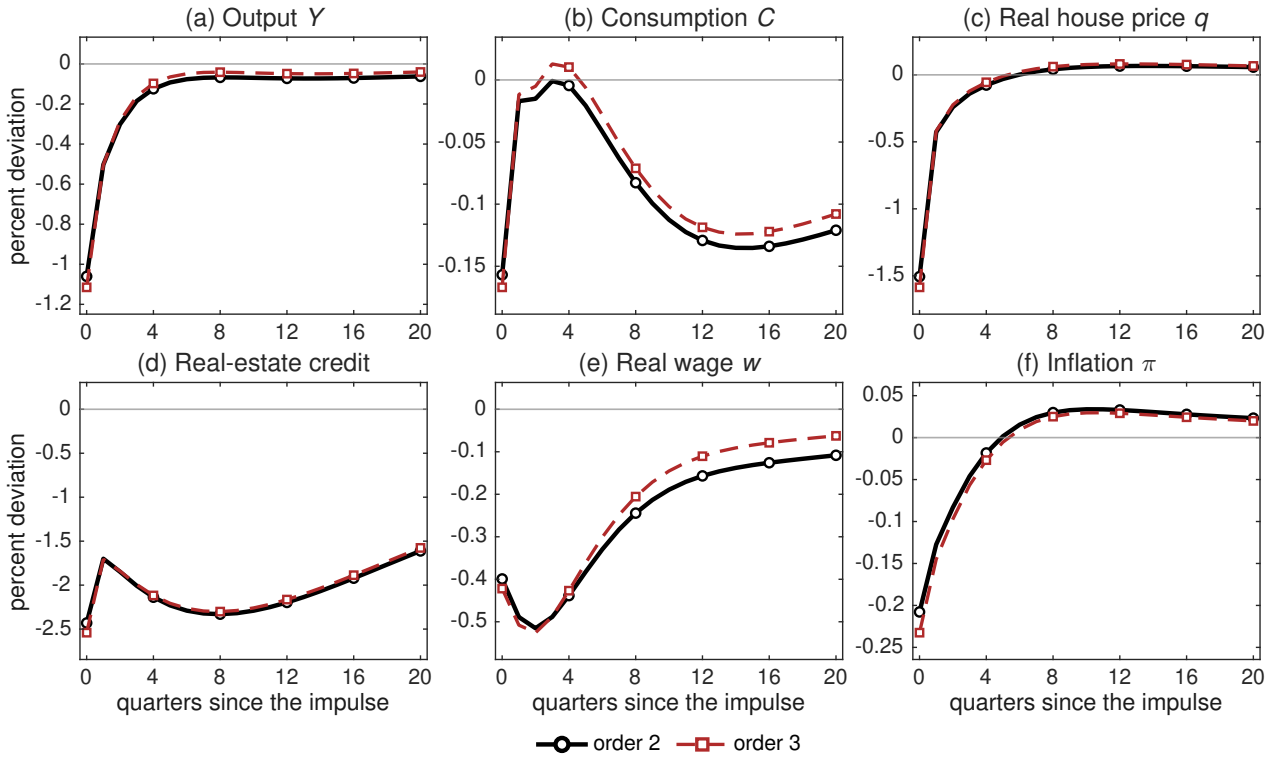
Notes. The measured disagreement path, inverted onto ν_t by Eqn. (26), and run through a second-order solve at $\chi_q = 8.59$ with productivity and policy shocks turned off. *Panel (a)* plots SLOOS net percent tightening on commercial real estate against the model's credit tightening in Eqn. (23). *Panels (b) and (c)* plot real-estate loans at all commercial banks and the real Case–Shiller national index, both deflated and entering as 100 log, against the model's credit and house price; both sides are Hodrick–Prescott filtered at $\lambda = 1600$. *Panel (d)* plots each survey's standardized D_t against the model's dispersion along the extracted path. Each model path starts from steady state in the quarter before its first observation.

Figure Appendix.5: Disagreement: Michigan vs. SCE



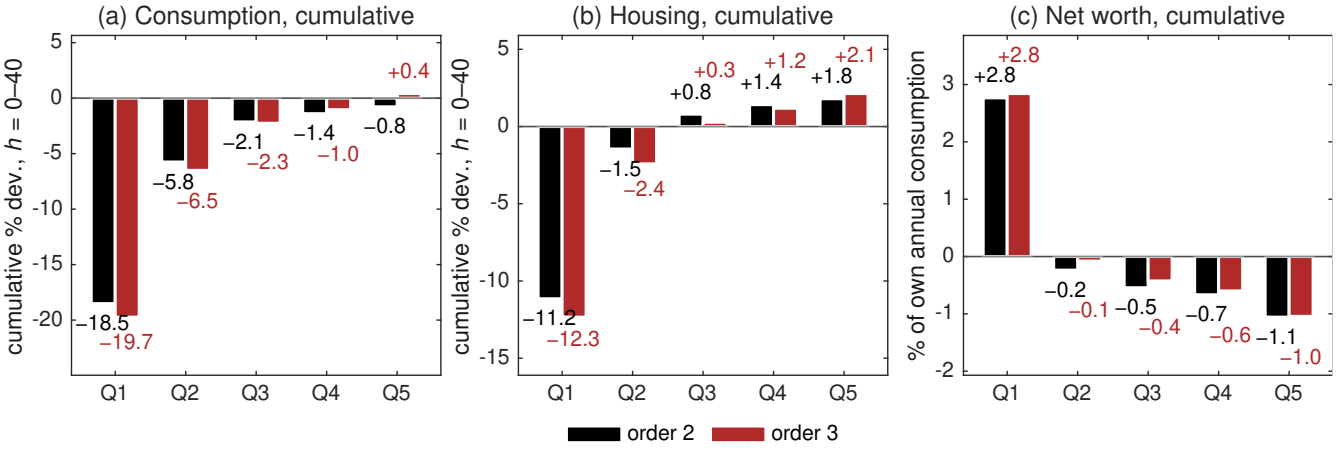
Notes. Panel (a) is the quarterly cross-sectional standard deviation of `HOMPX1`, winsorized 1/99 pooled and weighted by `WT`, using the usable Michigan responses, alongside the paper's SCE series over its own sample. The marked quarters are each series' maximum. The dotted line is 2024Q2, the quarter that includes both survey modes and in which the annual sample size doubles. Panel (b) plots the two series standardized over the quarters they share; the dashed line is the least-squares fit, and its slope equals the correlation because both axes are standardized.

Figure Appendix.6: Aggregate responses at perturbation orders two and three



Notes. Impulse responses to a $+1$ standard deviation innovation ν_t to Eqn. (10) in percent deviation from the steady state, simulated from the same decision rules at each order.

Figure Appendix.7: Distributional Effects: Second vs. Third Order



Notes. This figure replicates panels (d) and (e) of Figure 6 under second- and third-order perturbations; panel (c) cumulates net worth over $h = 0$ to 40 instead of discounting it as in panel (f) of Figure 6.